Confinement Width:
Interface Limits for Evaluator-Relative Control

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Abstract

For controlled dynamics $x_{t+1} = F(x_t, u_t)$ and an evaluator region $R_V = \{x : V(x) \geq \tau\}$, we study the cost of confinement: keeping closed-loop trajectories in R_V under initial-state uncertainty. We price controllers by a split read/write interface: s_r quantized sensing ports and s_w quantized actuation ports, each carrying b bits per step. Near a boundary equilibrium, the local-invariance problem with m unstable modes invokes classical results, the data-rate theorem and invariance entropy [65, 54, 14, 39], which force both the read rate and the declared finite write-branching rate to exceed $\sum_i \log_2 |\lambda_i|$ bits per step. Thus each side has width $\Omega(m/b)$, and $\Omega(m)$ for fixed port precision b, under uniform hyperbolicity. Description length inherits the bound only for loop-free interfaces, a restriction we state rather than hide. Critical-point statistics for Gaussian random fields [7, 5, 27] motivate the conjecture that evaluator regions in generic high-dimensional landscapes have equilibria in a collar of the boundary of extensive unstable index $m = \alpha N$: a null model, not a claim about trained networks. A transfer proposition shows that a d-dimensional harness with evaluator sufficiency and per-step measurement caps each read/write side at d, at sufficient port precision; a minimal exhibit shows the failure mode, an optimizing occupant routing evaluator-relevant change through the projection's kernel. [r3: The exhibit is now implemented in a companion repository, and the transfer proposition, its quantized and predictive variants, the fiber-ambiguity obstruction, and the discrete skeleton of the rate bound are machine-checked in Lean 4 without Mathlib.] Absent non-generic structure or a valid quarantine, low-width conditioning steers rather than confines.

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1 Motivation: from interpretability to maintained control

We wish to constrain the behavior of Large Language Models (LLMs) in agentic harnesses by applying a web of smaller models such as Tiny Recursive Models (TRMs) and Lattice Deduction Transformers (LDTs) [36, 69, 18, 24] to filter out misaligned tool calls and computer use trajectories from being realized. Such control architectures could render the alignment-problems and mechanistic interpretability problems of LLMs into a tractable security and control flow problem, in the sense of the trusted/untrusted protocol framing of AI control [33]. However, can a smarter model devise ways to exploit the filters and control membranes obstructing its goals? We explore complexity mathematics to quantify the problem.

The motivating problem is not merely whether a model can be made to pass an evaluator at one time. It is whether an evaluator's judgment can remain attached to the intended property while the judged system is trained, prompted, tooled, self-modified, or otherwise recursively improved. In that regime, alignment is less like classifying a finished artifact and more like maintaining an invariant of a changing dynamical system. The failure modes of a fixed proxy under optimization pressure have their own literature, under the names of Goodhart variants, reward gaming, and misaligned objectives [46, 61, 73]; the present note prices one of them, fiber routing, rather than cataloguing them.

This motivates the shift from static description to live control. A mechanistic explanation, a safety feature, a constitutional rule, or a steering vector can all be useful, but none of them is automatically a controller. To bear safety load, an interpretation must supply or induce a measured state through which evaluator-relevant facts continue to factor, a model of how that state evolves, and a feedback interface wide enough to correct departures. The question is therefore not just *what did we find inside the model?* but *what resource budget is required to keep the evaluator-relevant trajectory inside the region we care about?*

The note isolates one local resource: confinement width. If a system is near an evaluator boundary and the local target has unstable modes, classical data-rate and invariance-entropy results

say that holding the local set invariant requires a per-step information flow. This does not prove a new theorem of control theory; it converts known lower bounds into an evaluator-relative accounting useful for scalable oversight [13, 35, 6]. In particular, it explains why the natural description-length framing is too optimistic for general programmatic controllers: rate constrains the plant–controller pipe, not merely the program that processes the pipe.

The constructive side is dimensional quarantine. Rather than attempting to confine the full high-dimensional base system directly, a harness may project the system into a low-dimensional state $z = \phi(x)$, verify that the evaluator approximately factors through that state, and enforce invariance there. This is the formal role of TRM/LDT-style harnesses in the note: they do not eliminate the confinement problem, but they can reduce its measured dimension when the projection is evaluator-sufficient and dynamically closed enough.

This also gives a prototype for FRC-style complexity mathematics [23, 25]. Confinement width is the local lower-bound term; quarantine transfer is the constructive upper route; fiber ambiguity is the obstruction; and the fidelity half-life $t^*(\lambda)$ is the first empirical observable. The intended research program is to replace slogans about "alignment under recursive improvement" with curves and inequalities: roundwise width requirements, audit costs for maintaining sufficiency, and measured failure times as optimization pressure increases.

[r3: Revision 3 adds, to the October 10 text, an implemented exhibit (§7.1), a Lean 4 development for the elementary parts of the argument (Appendix E), a bits-before-ports accounting of the headline bound (Remark 2.A), a discrete skeleton of Proposition 3.1 (Lemma 3.A), a reconciliation with the October 10 revisions of the FRC papers (§9.1), and citations. The October 10 numbering is preserved; new items carry letters.]

2 Opening: what is borrowed, what is priced

The lower-bound mechanism in this note is not new, and we say so here rather than leaving it for a referee: the per-step information requirement for holding a set invariant is the data-rate theorem, with the discrete-time bits-per-step normalization closest to the Nair–Evans finite-rate stabilization line [54] and the earlier Wong–Brockett limited-information results [70], and, in its set-invariance form, invariance entropy (Colonius–Kawan) [14, 39, 38]. The feedback-entropy formulation of the same requirement is Nair, Evans, Mareels and Moran [55], and the relation between the two entropies is worked out in [15]. Tatikonda–Mitter and Sahai–Mitter are used here for the broader communication-constrained and noisy-channel program, where channel hypotheses and reliability notions matter [65, 59]; the surveys [56, 34] place both lines. Viability theory supplies the constrained-dynamics vocabulary, viable region, regulation map, and maintained constraint set, not the information-rate bound itself [4]. Earlier control work also treats realization burden as mathematical, for example through Brockett's minimum-attention viewpoint and later minimum-attention work, and through static coder-controller / lookup-table constructions in the invariance-entropy literature [9, 42, 21, 67].

What none of these price, and what this note defines and bounds, is a composite of three things. First, the read/write interface footprint of confinement: the number of quantized sensing channels and actuation channels a controller must use per step. The theorem reports the bottleneck lower bound, while the FRC accounting keeps read and write costs separate. Second, the behavior of that footprint under the critical-point statistics of generic high-dimensional landscapes, where boundary equilibria of evaluator regions conjecturally carry extensive unstable index, so the classical per-step requirement becomes extensive. Third, the maintenance cost of a low-dimensional quarantine: the validity of the projection and evaluator through which the width reduction is purchased, against

an occupant optimizing under the harness's own proxy. The abstraction and verification literatures [64, 30] do not price this because they typically assume the abstracted system is not trying to leave.

The claim ceiling, stated once and never exceeded: we compose and price known lower bounds in evaluator-relative AI systems; we do not invent invariance entropy.

2.1 Definitions

Dynamics and precision. Consider dynamics

$$x_{t+1} = F(x_t, u_t), \qquad x \in X \subseteq \mathbb{R}^N, \quad u \in U \subseteq \mathbb{R}^p. \tag{1}$$

Port values live on a fixed-precision grid $\delta\mathbb{Z} \cap [-M, M]$; the state x itself ranges over the continuum X.[1] A scalar port therefore carries

$$b = \log_2\left(\frac{2M}{\delta} + 1\right) \tag{2}$$

bits per step: $O(1)$ in N, and $O(\log(M/\delta))$ in precision. Where the landscape picture is wanted, take $F(x, 0) = x - \gamma\nabla L(x)$, so the uncontrolled Jacobian is $I - \gamma H(x)$ with H symmetric and GOE as the null model.

Evaluator region. Let $V : X \to \mathbb{R}$ be an evaluator and let

$$R_V = \{x : V(x) \geq \tau\}. \tag{3}$$

Ports and interfaces. A port is a scalar plant–controller channel exchanging one fixed-precision value per step: sensing ports carry readings in, actuation ports carry controls out. For a controller θ, let $s_r(\theta)$ and $s_w(\theta)$ count active sensing and actuation ports. Define the corresponding read and write rates

$$R_r(\theta) = b\, s_r(\theta), \qquad R_w(\theta) = b\, s_w(\theta). \tag{4}$$

We write

$$W_\wedge(\theta) = \min(s_r(\theta), s_w(\theta)), \quad W_\pm(\theta) = (s_r(\theta), s_w(\theta)), \quad W_\Sigma(\theta) = s_r(\theta) + s_w(\theta). \tag{5}$$

The quantity $W_\wedge$ is the bottleneck of the loop and is used for lower-bound corollaries. The pair $W_\pm$, or the scalar footprint W_Σ when a scalar is needed, is used for resource accounting. Density of the functional behind a port is irrelevant to width: a dense linear readout $w^\top x$ through one port is one port carrying b bits per step.

Assumption 2.1 (No side channels)**.** All plant–controller interaction passes through the priced ports. Without this, every bound below is void: a plant that broadcasts its state at unbounded precision outside the accounting supports zero-width confinement trivially. The theorem prices the declared interface; undeclared channels are not denied by the theorem, but treated as exactly the kind of unpriced leakage that the fiber-ambiguity and audit sections are meant to expose. [r3: The same accounting problem, side information defeating a performance bound unless it is itself priced, is treated for disturbance attenuation by Martins, Dahleh and Doyle [49].]

[1] The July wording, "coordinates live on a fixed-precision grid," allowed the reading that the state is quantized. Then X is finite, a collar has zero Lebesgue measure, Assumption 3.1 cannot hold, and the asymptotic rate of Theorem 3.1 is zero. The grid constrains what crosses the interface, not where the plant can be.

Assumption 2.2 (Noiseless quantized ports)**.** Ports are deterministic digital channels at b bits per plant step. This noiseless quantized-interface model is the normalization used by the width theorem. Noisy channels are strictly harder for the controller and require explicit channel hypotheses; in that setting, anytime capacity rather than raw Shannon capacity is the operative stabilization notion [65, 59], and is out of scope here. [r3: Bode-type limitations over noisy channels are treated in [48].]

Definition 2.1 (Confinement width and description length)**.** Over controllers satisfying Assumptions 2.1 and 2.2 that keep every closed-loop trajectory from the stated uncertainty class in R_V for all $t \le T$ with failure probability at most ϵ, define the bottleneck and scalar-footprint confinement widths

$$W_V^{\wedge}(R,T,\epsilon) = \min_{\theta} W_{\wedge}(\theta), \quad W_V^{\Sigma}(R,T,\epsilon) = \min_{\theta} W_{\Sigma}(\theta), \quad L_V(R,T,\epsilon) = \min_{\theta} |\theta|. \tag{6}$$

When read and write costs should remain non-scalar, use the pair-valued footprint $W_{\pm}(\theta)$ and the corresponding Pareto frontier. Here $|\theta|$ is description length, equivalently $\|\theta\|_0$ up to logarithmic addressing in fixed precision [53]. Where the old name is wanted, set $C_V = (W_V^{\wedge}, W_V^{\Sigma}, L_V)$. The theorem below attaches to the read and write rates separately, with $W_V^{\wedge}$ as the bottleneck corollary. The quantity L_V is bounded only under the restricted interface class in Remark 3.4.

The uncertainty class is a parameter of the definition and must be nondegenerate: a viable collar of positive volume in the unstable coordinates, or a persistent disturbance of positive amplitude. Degenerate classes admit zero-width open-loop confinement.

Remark 2.1 (What the state space is)*.* Nothing in this note requires literal confinement in parameter space. X is any state space on which the evaluator region and the dynamics are jointly defined: parameter state under training updates, behavior or activation state under inference dynamics, harness state under a projection, or deployment-trajectory state under an operational semantics. Conditioning, prompts, fine-tunes, steering, reshapes F rather than pinning x. The width question is posed after the reshaping: given the dynamics an intervention leaves behind, how many quantized channels of online observation and actuation does it take to keep evaluator-relevant trajectories inside R_V? A zero-width intervention can relocate equilibria, which is steering and may be sufficient; what it cannot do, near boundary equilibria satisfying the hypotheses below, is supply the per-step bits that Theorem 3.1 says holding requires.

Remark 2.2 (Competence floor)*.* Remark 2.1 concedes that a zero-width intervention relocating equilibria into the interior of R_V may be sufficient. Where it is, confinement costs nothing and the width question is empty. This is the counterpart, in the present frame, of the agent that satisfies a shutdown property by never pursuing a goal (Remark 10.2). The width bound bites only if the system is required to operate near ∂R_V, and a performance requirement is what supplies that: for a task-performance functional perf and floor η_c (written η in [23]; the subscript keeps it apart from the closure defect η_t of §5), replace R_V by $R_V \cap \{x : \mathrm{perf}(x) \ge \eta_c\}$, which is again an evaluator region, that of $V' = \min(V - \tau, \mathrm{perf} - \eta_c)$ at level 0, though a non-smooth one that falls outside the $V = g(L)$ setting of Conjecture 4.1. When performance and evaluator margin trade off, the retreat to the interior is excluded. That they do trade off is a hypothesis this note has been using without stating, and it is not derived here. A reward floor on the task family is the light choice of perf; Goertzel's pragmatic general intelligence [31] is a heavier one, relative to chosen distributions over environments and goals and defined only where its sum converges.

Remark 2.A (Bits before ports)*.* [r3: new.] The quantity the theorem controls is a rate in bits per step, $h_2(A_u) = \sum_i \log_2 |\lambda_i|$; the port count is that rate divided by the port precision b, and b is large for the channels an engineer first thinks of. Table 1 evaluates $\lceil m \log_2(1+\delta)/b \rceil$ for m

unstable modes at a uniform gap δ (Assumption 3.2), which is the structured case; in the null model of Remark 4.3 there is no uniform gap and the rate is $Nc(\gamma)$ with $c(\gamma) \approx 4\gamma\sigma/(3\pi \ln 2)$, about 0.03 bits per dimension per step at $\gamma\sigma = 0.05$. Either way the per-mode rate is a fraction of a bit, and at $b = 32$ (a float port) one port covers of the order of a hundred weakly unstable modes. The $\Omega(m)$ reading is therefore correct only for fixed low b, and the channels with low b are: a human approval or rejection (one to a few bits per step), a token stream under a short budget ($b \approx \log_2 |\text{vocab}| \approx 17$ bits per token, with few tokens per decision), and a symbolic action alphabet such as the four registry operations of the companion harness [22] (two bits per step). Those are the channels through which scalable oversight is proposed to run [13, 6, 11]. The bound is weak where the loop is a float pipe and strong where the loop is a person or a grammar. Remark 6.2 is the same observation from the other side: at coarse b the cap of Corollary 6.1 rises to $d\lceil b'/b \rceil$.

m	δ	h_2 (bits/step)	$b = 1$	$b = 3$	$b = 17$	$b = 32$
100	0.01	1.4	2	1	1	1
100	0.1	13.8	14	5	1	1
100	1	100	100	34	6	4
100	10	346	346	116	21	11
1000	0.01	14.4	15	5	1	1
1000	0.1	137.5	138	46	9	5
1000	1	1000	1000	334	59	32
1000	10	3459	3460	1154	204	109

Table 1: [r3: new.] Ports per side required by Theorem 3.1 at uniform gap δ, i.e. $\lceil m \log_2(1+\delta)/b \rceil$. The column $b = 1$ is a binary verdict channel; $b = 3$ a coarse human grade; $b = 17$ one token of a 10^5-word vocabulary; $b = 32$ a float port.

3 The local lower bound

The load-bearing statement in this section is the linear local-invariance problem. Work in local coordinates with $x^* = 0$, fix a precompact coordinate neighborhood O of 0, and consider the declared-interface plant

$$x_{t+1} = Ax_t + Bu_t. \tag{7}$$

For a nonlinear system one may set $A = D_x F(x^*, 0)$ at a boundary equilibrium $x^* \in \partial R_V$, but the nonlinear transfer of the entropy bound requires the usual geometric hypotheses and is not the theorem stated here. Let E^u be the unstable subspace of A, let P_u be the spectral projection onto E^u along an A-invariant complementary subspace, and write $m = \dim E^u$.

Definition 3.1 (Viable local collar)**.** A viable local collar is a compact set

$$K_0 \subset R_V \cap O \tag{8}$$

whose projection $P_u(K_0)$ has positive m-dimensional Lebesgue measure. Let $A_u = A|_{E^u}$ and define

$$h_2(A_u) = \log_2 |\det A_u| = \sum_{\lambda_i \in \sigma(A_u)} \log_2 |\lambda_i|. \tag{9}$$

Assumption 3.1 (Nondegenerate viable uncertainty)**.** The confinement guarantee is quantified over a viable local collar K_0 in the sense of Definition 3.1, equivalently over persistent disturbances whose reachable uncertainty has positive P_u-volume in the unstable coordinates. This is what generates the bits.

Assumption 3.2 (Uniform unstable hyperbolicity)**.** Every eigenvalue of A_u satisfies $|\lambda_i| \geq 1 + \delta$ for some $\delta > 0$.

Lemma 3.1 (Mode coverage: PBH necessity)**.** *Any controller, static or dynamic and of any width, that locally confines under Assumptions 3.1 and 3.2 must have every unstable mode in E^u controllable through its actuation ports and observable through its sensing ports, in the PBH sense.*

Sketch. An uncontrollable or unobservable unstable mode evolves autonomously in the closed loop by the Kalman decomposition and eventually leaves the bounded local target in the P_u coordinate, so it cannot be guaranteed to stay in $R_V \cap O$. This constrains which subspaces the interface touches, not yet how many ports it needs. □

Remark 3.1 (First public correction: rank counting fails)*.* An earlier draft asserted that stabilizing m unstable modes forces feedback of rank at least m, whence $\|\theta\|_0 \geq m$. This is false. For a controllable single-input pair (A, g), pole placement assigns the closed-loop spectrum arbitrarily, so a rank-one perturbation $gk^\top$ can stabilize any number of unstable modes. For non-normal matrices, low-rank corrections can move unboundedly many eigenvalues. The counterexample supplies no cheap controller in the present accounting, the placing gain is dense and exact, but the obstruction that survives is informational, not algebraic.

PBH mode coverage says which unstable subspaces must be touched; it does not count ports. The port count comes from the information rate required to refine the uncertainty set fast enough to offset unstable expansion.

Theorem 3.1 (Split read/write local-invariance width lower bound; linear local form)**.** *Under Assumptions 2.1, 2.2, 3.1 and 3.2, any coder-controller keeping every trajectory from K_0 in $R_V \cap O$ for all t must satisfy*

$$\liminf_{T\to\infty} \frac{B_{\mathrm{read}}(T)}{T} \geq h_2(A_u), \tag{10}$$

$$\liminf_{T\to\infty} \frac{B_{\mathrm{write}}(T)}{T} \geq h_2(A_u), \tag{11}$$

where $B_{\mathrm{read}}(T)$ and $B_{\mathrm{write}}(T)$ are the total read and write bits used up to time T. Consequently, for fixed per-step port counts,

$$b\, s_r(\theta) \geq h_2(A_u), \qquad b\, s_w(\theta) \geq h_2(A_u). \tag{12}$$

Thus the bottleneck width obeys

$$W_\wedge(\theta) \geq \frac{h_2(A_u)}{b} \geq \frac{\log_2(1+\delta)}{b}\, m. \tag{13}$$

Each side of the declared interface therefore has width $\Omega(m/b)$. For fixed port precision b and fixed hyperbolicity gap δ, each side has width $\Omega(m)$.

Sketch. The read-side inequality is the discrete-time data-rate theorem in the finite-feedback-rate lineage [54]: the controller's observations must refine the viable uncertainty set quickly enough to offset expansion by A_u. The write-side inequality is the corresponding invariance-entropy / spanning-control-function count [14, 39]: with only s_w quantized actuation ports, the controller can realize at most $2^{s_w b}$ distinct control branches per step, so the admissible control functions cannot cover the expanding unstable uncertainty unless the write alphabet grows at the same

exponential rate. The volume-counting intuition is that the uncontrolled dynamics expand E^u volumes by $|\det A_u|$ per step; sensing partitions the uncertainty set into at most $2^{s_r b}$ cells per step, and actuation branches the response into at most $2^{s_w b}$ controls per step. Invariance requires each refinement to offset that expansion. Hence both inequalities in (12), and the bottleneck corollary (13). The theorem prices the declared finite digital interface; unpriced analog or side channels violate Assumption 2.1 rather than refute the bound. □

Proposition 3.1 (Finite-horizon form, with the constant)**.** *Let the plant be affine, $x_{t+1} = Ax_t + Bu_t + f_0$, under Assumptions 2.1, 2.2 and 3.1, and let vol_u denote m-dimensional Lebesgue outer measure of the P_u-projection of a set. For a deterministic coder-controller let N_T be the number of distinct control sequences $u_{0:T-1}$ it applies as x_0 ranges over K_0. If every trajectory from K_0 lies in $R_V \cap O$ at time T, then*

$$\log_2 N_T \geq T h_2(A_u) - c_0, \qquad c_0 = \log_2 \frac{\mathrm{vol}_u(R_V \cap O)}{\mathrm{vol}_u(K_0)} \geq 0. \tag{FH}$$

Since the control sequence is a function of the write symbols, which are a function of the read symbols, $\log_2 N_T \leq \min\{B_{\mathrm{read}}(T), B_{\mathrm{write}}(T)\}$, and for fixed per-step port counts

$$b\, s_r \geq h_2(A_u) - \frac{c_0}{T}, \qquad b\, s_w \geq h_2(A_u) - \frac{c_0}{T} \qquad \text{for every } T \geq 1 \text{ at which confinement holds.}$$

Here E^u is the generalized eigenspace of the eigenvalues with $|\lambda| > 1$, the plant is noise-free, symbols are of fixed length on a port schedule that does not depend on the data, and the controller's initial state does not depend on x_0. Theorem 3.1 is the limit $T \to \infty$; Assumption 3.2 is used only for the last inequality in (13).

Proof. Partition K_0 into the cells C_i, $i \leq N_T$, of initial states that receive the i-th control sequence. On C_i the control sequence is fixed, so $x_T = A^T x_0 + g_i$ for a vector g_i depending only on i (it collects the controls and the drift f_0). Since P_u commutes with A, $P_u x_T = A_u^T P_u x_0 + P_u g_i$, so $A_u^T P_u(C_i) + P_u g_i \subseteq P_u(R_V \cap O)$ and

$$|\det A_u|^T \,\mathrm{vol}_u(C_i) \leq \mathrm{vol}_u(R_V \cap O),$$

which is finite because O is precompact. Summing over cells, by subadditivity,

$$\mathrm{vol}_u(K_0) \leq N_T |\det A_u|^{-T} \,\mathrm{vol}_u(R_V \cap O).$$

Take logarithms. □

This is the classical counting argument behind [54, 14, 39] written for the split interface; nothing in it is new. It supplies the constant that Remark 3.3 asked for. It covers the collar form of Assumption 3.1 and not the persistent-disturbance alternative.

Lemma 3.A (Discrete skeleton of (FH); machine-checked). [r3: new.] Replace Lebesgue measure by cardinality. Let the controller's uncertainty set at step t have u_t elements, let the plant branch over a disturbance alphabet of size Λ injectively in (state, disturbance) for each fixed control, and let the controller observe one of $A = 2^R$ symbols per step and choose its control from that symbol alone. Then

$$u_{t+1} A \geq \Lambda u_t, \qquad \text{hence} \qquad u_T A^T \geq \Lambda^T u_0, \tag{D1}$$

which in logarithms reads $TR \geq T \log_2 \Lambda - \log_2(u_T/u_0)$: the shape of (FH) with $\log_2 \Lambda$ for h_2 and $\log_2(K/u_0)$ for c_0 when the target has K elements. If $u_t \leq K$ for all t then $\Lambda \leq A$, i.e. $R \geq \log_2 \Lambda$.

Proof. By pigeonhole some observation symbol is received by at least u_t/A states of the uncertainty set. On that class the control is constant, so by injectivity the Λ disturbance branches of those states are $\Lambda\lceil u_t/A\rceil$ distinct next states, all consistent with the controller's information. Hence $u_{t+1} \geq \Lambda u_t/A$. Writing $w_t = u_t A^t$ gives $w_{t+1} \geq \Lambda w_t$, so $w_T \geq \Lambda^T u_0$. If $\Lambda \geq A+1$ and $u_t \leq K$ for all t, take $T = KA+1$: by the Bernoulli inequality $(A+1)^T \geq A^T + TA^{T-1} = (K+1)A^T + A^{KA} > KA^T \geq u_T A^T = w_T \geq \Lambda^T \geq (A+1)^T$, a contradiction. The step from the recurrence to (D1) is `weighted_growth` and the contradiction is `data_rate_necessity` in Appendix E; the pigeonhole step is stated there as a hypothesis and proved only here. Proposition 3.1 is the same count with cells of positive measure in place of singletons; the Lean development does not reach it because core Lean has no Lebesgue measure. □

Remark 3.2 (Boundary-transversal pricing is not claimed here). Because Theorem 3.1 requires trajectories to remain in $R_V \cap O$, it prices local invariance in the coordinate chart. A trajectory that remains in R_V while sliding along ∂R_V and exiting O still violates this local target, so boundary-tangent unstable modes are included in $m = \dim E^u$. In applications where exit through $\partial O \cap R_V$ is operationally acceptable, the relevant count should be the boundary-transversal evaluator-failure modes rather than the full unstable index. Formalizing that quotient or exit problem requires a separate viability / invariance-entropy lemma and is left to future work.

Remark 3.3 (Finite horizon). Over finite T the controller may coast on the margin between the viable collar K_0 and the boundary of $R_V \cap O$, paying an additive number of bits determined by the initial unstable-uncertainty volume and the available safety margin. The clean statement here is asymptotic; a circulation draft should either write the finite-horizon correction explicitly or avoid relying on it. It is now written: Proposition 3.1, with constant c_0, and Proposition 10.1 uses it.

Remark 3.4 (Second public correction: description length does not bound width). It is tempting to conclude $\|\theta\|_0 \geq W_\wedge(\theta)$, or even $\|\theta\|_0 \gtrsim W_\Sigma(\theta)$, but this is false for general programmatic controllers. A looped fixed-precision program can process arbitrarily many ports with $O(1)$ nonzero parameters by reading them through generated addresses and accumulating. Rate constrains the pipe, not the program. Thus L_V inherits the $\Omega(m/b)$ bound, and the $\Omega(m)$ bound for fixed b, only under a restricted class:

Loop-free/static interfaces: read and write maps are fixed sparse linear or affine maps with no address generation, so each active port consumes at least one dedicated nonzero parameter. Then $\|\theta\|_0$ is at least the active interface footprint up to addressing constants, and L_V inherits the split width lower bound. For general programmatic controllers, only the read/write width bound stands.

Remark 3.5 (Affine plants: rate and authority are separate prices). Proposition 3.1 holds with a drift term f_0 because a constant translates images and leaves volumes alone. The point x^* therefore need not be an equilibrium of the uncontrolled dynamics for the rate bound to apply: with $A = D_x F(x^*, 0)$, what is counted is the unstable index of the Jacobian at a boundary point, equilibrium or not. What a drift adds is a second, different requirement. If a trajectory stays in the bounded set O for all t, the time averages of $x_{t+1} - x_t$ tend to zero, so f_0 must lie within $\|A - I\| \operatorname{rad}(O)$ of the closure of $-B \operatorname{conv}(U)$, where $\operatorname{rad}(O) = \sup_{x \in O} \|x\|$: the actuators must be able to cancel the drift. That is a condition on the range of B and the size of U, priced in actuation amplitude and not in bits. A low-rate, high-authority push is the steering of Remark 2.1; it does not discharge the rate bound at the point it steers to. For a nonlinear F the affine model at a non-equilibrium point is valid only while the trajectory stays in O, and the transfer remains unproved, as stated after (7).

Remark 3.6 (Where ϵ and T of Definition 2.1 went). Definition 2.1 allows a failure probability ϵ and a finite horizon T; Theorem 3.1 is the case $\epsilon = 0$, all T, and (FH) is its finite-T form. For

$\epsilon > 0$, let x_0 be drawn from a law on K_0 whose P_u-marginal has density at most $\rho_{\max}$. The same count, applied to the set of initial states that are confined at time T, gives

$$\log_2 N_T \geq T h_2(A_u) - \log_2 \frac{\rho_{\max} \operatorname{vol}_u(R_V \cap O)}{1-\epsilon},$$

for deterministic coder-controllers; for randomized ones with a fixed bit budget the same right side bounds the budget, by averaging over private randomness. A positive ϵ moves the constant and leaves the rate where it was.

4 The generic extension

Conjecture 4.1 (Extensive local unstable index)**.** *For landscapes in the Gaussian-random-field class, such as spherical-spin-glass ensembles, and evaluators tied to the landscape, $V = g(L)$ with g monotone, equilibria in a collar $\{|V - \tau| \leq \epsilon_c\}$ of the boundary of evaluator regions at non-extreme levels carry extensive local unstable index $m = \dim E^u = \alpha N$ with high probability. This would follow from Kac–Rice complexity asymptotics restricted to a thin shell around the level set $\{V = \tau\}$, together with the level–index tradeoff of Bray–Dean [7] and random-matrix large-deviation suppression of atypically one-signed spectra [20]. Consequently, under Theorem 3.1, both declared read and write widths are $\Omega(N/b)$ in the null model, and $\Omega(N)$ for fixed b, with a constant set by the step size γ (Remark 4.3).*

The restriction of Kac–Rice to an evaluator level set is an unproved step and is named as such [7, 5]; [r3: the second-moment technology that would be needed is the one developed for spherical p-spin complexity [63, 28].] A numerical check is specified as evidence rather than theorem: sample Gaussian fields at $N = 50$–200, locate boundary equilibria of a level set, and histogram the local unstable index. The sharper evaluator-boundary question that subtracts tangent exit directions is the boundary-transversal problem isolated in Remark 3.2.

Revision note (October 2026). Two changes to the July statement. (i) For a fixed τ an equilibrium lying exactly on $\{V = \tau\}$ is non-generic. The landscape L is almost surely a Morse function; on a compact domain it has finitely many critical points, these take finitely many values of V, and all but finitely many levels τ carry none. The conjecture is therefore about equilibria in a collar of the boundary, and "boundary equilibrium" elsewhere in the note should be read that way. Nothing is lost: Proposition 3.1 uses the linear plant on O and a collar $K_0 \subset R_V \cap O$, and never that x^* lies on ∂R_V. (ii) The Bray–Dean tradeoff relates the index of a critical point to the level of the field itself. It bears on $\{V = \tau\}$ when V is a monotone function of L, so that the boundary is a level set of the landscape. For an evaluator not tied to L the level-set restriction is the whole of the unproved step, and the index of collar equilibria is that of whatever critical points the dynamics reaches, which this note does not characterize.

GOE/Gaussian genericity is a null model. Trained networks are structured, low-rank, modular, with bulk-zero-plus-outlier Hessian spectra that already depart from GOE, and no claim is made that they are generic fields. The conjecture fixes the default against which structure must be exhibited: a confinement proposal either identifies the non-generic structure it exploits, supplies a quarantine in the sense of section 5, or inherits the $\Omega(N/b)$ local read/write floor, which is $\Omega(N)$ for fixed b.

Corollary 4.1 (Conditioning gap, conditional)**.** *Price standard interventions by the declared read-/write interface. A prompt or system message writes no runtime port into the evolving state after*

$t = 0$. *Equivalently, prompts and system messages are treated here as reshaping F before the confinement problem is posed, not as priced runtime feedback channels; see Remark 2.1. Offline preference training has zero runtime rate by definition. Activation steering along r fixed directions [68, 43, 74] is r write ports, or rb bits per step. If the operating regime is near boundary equilibria satisfying Conjecture 4.1, then no controller with $o(N/b)$ declared read or write width meets the requirement of Theorem 3.1, and no $o(N)$ declared read or write width does for fixed b. Stated at its ceiling: absent exploitable non-generic structure or a valid low-dimensional quarantine, low-complexity conditioning should be expected to steer rather than confine.*

Remark 4.1 (Methodological parallel, non-load-bearing). The same critical-point statistics, run on random supergravity potentials, led string cosmology to conclude that metastable perches are exponentially rare and the generic picture is a slow roll along a gradient floor [47]. The parallel carries no evidential weight here; it earns its three sentences only by showing the inversion, perch statistics read as a statement about residence rather than search, has been executed before.

Remark 4.2 (Conjectural energy and fragility obstruction). Earlier drafts carried a Gramian-conditioning argument that bounded-magnitude fixed-precision controllers need actuation support $\Omega(m)$. Two gaps keep it out of the main text: pole placement is sufficient rather than necessary, and the cited energy-scaling results concern reaching target states in network systems rather than gain magnitudes for saddle confinement under GOE spacing. If retained, this material should be re-founded on the discrete-time Bode sensitivity integral [51, 26] and quantized stabilization, then stated only as a fragility conjecture complementing the width bound.

Remark 4.3 (The null model violates Assumption 3.2; extensivity survives through h_2). With $F(x,0) = x - \gamma\nabla L(x)$ and $\gamma\|H\| < 2$, the unstable multipliers of $I - \gamma H$ are $1 + \gamma|\lambda_i|$ over the negative eigenvalues λ_i of H. A GOE-type spectrum has density through zero, so these multipliers accumulate at 1 and no gap $\delta > 0$ is uniform in N: Assumption 3.2 fails in the very model the conjecture is stated for, and the last inequality of (13) is unavailable. The first inequality of (13) does not use the gap, and gives

$$h_2(A_u) = \sum_{\lambda_i<0} \log_2\bigl(1+\gamma|\lambda_i|\bigr) = N\bigl(c(\gamma)+o(1)\bigr), \qquad c(\gamma) = \int_{\lambda<0} \log_2\bigl(1+\gamma|\lambda|\bigr)\, d\rho(\lambda),$$

with ρ the limiting spectral density of the Hessian at the equilibria in question, a shifted semicircle in the spin-glass ensembles [5]. For the centered semicircle of radius 2σ, $c(\gamma) \geq \bigl(4\gamma\sigma/(3\pi) - \gamma^2\sigma^2/4\bigr)/\ln 2$, and $c(\gamma) \approx 4\gamma\sigma/(3\pi\ln 2)$ when $\gamma\sigma \ll 1$. The null-model width is thus $\Omega(\gamma\sigma N/b)$: extensive in N, with a constant proportional to the step size. The form $m\log_2(1+\delta)/b$ would have suggested a constant independent of γ, and that is not available here.

Remark 4.A (Which state space the conjecture lives in). [r3: new.] The conjecture is a statement about critical points of a scalar field on parameter space, in the lineage that carried spin-glass complexity into deep learning [17, 12]. Three things follow that the text leaves implicit. (i) The sentence "bulk-zero-plus-outlier Hessian spectra" above has measurements behind it: at convergence the Hessian of a trained network has a bulk concentrated near zero and a number of outliers of the order of the class count [58, 29, 57]. In parameter space the unstable index of a local target is therefore small, not extensive, and Remark 4.3's $c(\gamma)$ is the wrong null there. (ii) The state spaces on which harnesses operate (Remark 2.1: activation, behavior, registry, trajectory state) have no GOE null at all; nobody has written the Kac–Rice problem for them, and it is not obvious the critical-point picture is the right one for an operational semantics. The partial-observability framing of the harness problem, where the right object is a belief state rather than a point, is the POMDP one [37]; whether the width accounting survives that translation is open. (iii) Corollary

4.1 is therefore conditional on a conjecture in a space the harness does not observe. It is retained because it states the default, and the abstract's last sentence should be read with that condition attached.

5 Dimensional quarantine

A projection $\phi : X \to Z$, with $\dim Z = d \ll N$, defines a harness state. Here LDT, expanded in §1 as Lattice Deduction Transformer, is glossed by the role it plays in this note, as *latent-dynamics tracker*: a harness maintaining an explicit low-dimensional state z with its own transition model. TRM-routed adapters instantiate the class. [r3: In the companion architecture paper the same role is played by the explicit lattice state and typed membrane of the TRM/LDT hybrid [24, 18]. The construction is an abstraction in the sense of the symbolic-control literature [64, 30], with the difference, priced below, that the abstracted system is optimizing against the abstraction.]

Definition 5.1 (Evaluator sufficiency)**.** A projection ϕ is (V, ϵ)-sufficient on a set $S \subseteq X$ if there exists $v : Z \to \mathbb{R}$ such that

$$|V(x) - v(\phi(x))| \leq \epsilon \qquad \text{for all } x \in S. \tag{14}$$

To avoid the circularity of "sufficient on the reachable set," S is required to be a-priori checkable and forward-adequate, for example $S \supseteq R_V \cup F(R_V \times U)$.

Definition 5.2 (Quarantined dynamics and policy-indexed defect)**.** Projected dynamics are written

$$z_{t+1} = G(z_t, u_t) + \eta_t, \tag{15}$$

where the closure defect is defined by

$$\eta_t := \phi(F(x_t, u_t)) - G(\phi(x_t), u_t). \tag{16}$$

This is the projected shadow of unmodeled fiber dynamics, a Mori–Zwanzig-style memory term [75, 52, 45], and is therefore statistically dependent on whatever the occupant's optimization touches in the fiber. Accordingly the defect bound is worst-case over the occupant's policy class and round-indexed:

$$\bar{\eta}(\Pi_k) = \sup_{\pi \in \Pi_k} \sup_{x \in S,\, u \in U} \|\phi(F(x, u)) - G(\phi(x), u)\|. \tag{17}$$

Optimization enlarges Π_k, so $\bar{\eta}$ is a function of round; guarantees below hold per round at the current $\bar{\eta}$. [r3: The Koopman view of the same projection, a linear model of a nonlinear system in a chosen observable coordinate [41, 50, 10], is the natural source of candidate G; its closure defect is what the Mori–Zwanzig memory term measures.]

6 Reactive transfer

Proposition 6.1 (Reactive quarantined confinement)**.** *Assume ϕ is (V, ϵ)-sufficient on S, v is L_v-Lipschitz, per-step defects satisfy $\|\eta_t\| \leq \bar{\eta}$, the controller measures $z_t = \phi(x_t)$ at every step, and the robust one-step condition holds:*

$$\forall z \text{ with } v(z) \geq \tau + \epsilon, \quad \exists u \in U \text{ such that } v(G(z, u)) \geq \tau + \epsilon + L_v \bar{\eta}. \tag{18}$$

If $v(\phi(x_0)) \geq \tau + \epsilon$ and $x_0 \in S$, then $x_t \in R_V$ for all t.

Proof. Induct on t. Since $x_t \in R_V \subseteq S$, forward adequacy gives $x_{t+1} \in F(R_V \times U) \subseteq S$, so sufficiency applies. From $v(z_t) \geq \tau + \epsilon$, choose u_t according to (18). Then

$$v(z_{t+1}) \geq v(G(z_t, u_t)) - L_v \|\eta_t\| \geq \tau + \epsilon. \tag{19}$$

By evaluator sufficiency, $V(x_{t+1}) \geq v(z_{t+1}) - \epsilon \geq \tau$. The base case is identical. □

The robust one-step condition is a discrete-time control-barrier-function condition [2, 40, 3]. Its verification is a d-dimensional problem, which is the point.

Remark 6.1 (Predictive variant). If z is measured only at $t = 0$ and the controller certifies a model trajectory $\hat{z}_{t+1} = G(\hat{z}_t, u_t)$, the margin inflates from $\epsilon + L_v\bar{\eta}$ to

$$\epsilon + L_v D_T, \qquad D_T = \bar{\eta}\,\frac{L_G^T - 1}{L_G - 1}, \tag{20}$$

with $D_T = \bar{\eta}T$ when $L_G = 1$. This compounds geometrically unless G is contractive on the relevant region. Predictive quarantine is therefore a short-horizon instrument; the design criteria are to build G contractive and measure ϕ online.

Corollary 6.1 (Width cap and measurement)**.** *Under Proposition 6.1, the confinement problem the controller actually solves lives in Z. The width bound applies to the local unstable index m_G of G for the projected local-invariance problem, with $m_G \leq d$, not necessarily to the full base dynamics, provided the evaluator factors through ϕ. A projected controller that reads and writes the full projected state has $s_r, s_w \leq d$, hence $W_\wedge \leq d$ and $W_\Sigma \leq 2d$ under the scalar-footprint accounting, provided one b-bit port per coordinate meets the quantized margin of Remark 6.2; at coarser b the cap is $d\lceil b'/b \rceil$ per side, with b' the per-coordinate precision that does. TRM/LDT-style harnesses do not magically solve confinement; they reduce and measure the dimensionality of the confinement problem. What is not reduced is the maintenance of (V, ϵ)-sufficiency and closure against an optimizing occupant: $\bar{\eta}(\Pi_k)$ and the validity of S are round-indexed quantities not bounded by d.*

Remark 6.2 (Quantized measurement: the margin it consumes). Proposition 6.1 assumes z_t is measured exactly, and the port accounting delivers it on the grid. Let $\hat{z}_t$ be the reading, with $\|\hat{z}_t - z_t\| \leq q$, where $q = (\delta/2)\sqrt{d}$ in the Euclidean norm for one b-bit port per coordinate, δ being the grid spacing of (2), with no coordinate saturating at $\pm M$; and let v and $G(\cdot, u)$ be Lipschitz, with constants L_v and L_G uniform in u, on the q-neighbourhood in which readings land. Suppose the one-step condition holds in the tightened form

$$\forall \hat{z} \text{ with } v(\hat{z}) \geq \tau + \epsilon - L_v q, \quad \exists u \in U_b \text{ such that } v(G(\hat{z}, u)) \geq \tau + \epsilon + L_v\bar{\eta} + L_v L_G q, \tag{18′}$$

with $U_b \subseteq U$ the controls the write ports can express. Then the conclusion of Proposition 6.1 holds for the controller that acts on $\hat{z}_t$. Indeed $v(z_t) \geq \tau + \epsilon$ gives $v(\hat{z}_t) \geq \tau + \epsilon - L_v q$, so (18′) supplies u_t, and

$$v(z_{t+1}) \geq v(G(z_t, u_t)) - L_v\bar{\eta} \geq v(G(\hat{z}_t, u_t)) - L_v L_G q - L_v\bar{\eta} \geq \tau + \epsilon.$$

Quantization therefore consumes margin $L_v(1 + L_G)q$, which enters (30). It also bounds the cap of Corollary 6.1 from below. Theorem 3.1 applied in Z gives $b\,s_r \geq h_2$ of the unstable part of the projected dynamics, so $s_r \leq d$ is attainable only if $b\,d$ is at least that h_2: a projected mode that expands faster than 2^b per step needs more than one b-bit port.

Remark 6.A (Formalization of Proposition 6.1 and Remarks 6.1, 6.2). [r3: new.] The proof of Proposition 6.1 uses the Lipschitz and defect hypotheses only through the single inequality

$v(G(\phi x, u)) \le v(\phi(F(x,u))) + c$ with $c = L_v\bar{\eta}$, and uses only the lower half of (14). The Lean statement `reactive_confinement` (Appendix E) is stated at that generality, over any value type with a transitive order and right-cancellative addition, with the existential in (18) realized by a fixed policy $\pi : Z \to U$. It concludes the three invariants $x_t \in S$, $v(\phi x_t) \ge \tau + \epsilon$ and $V(x_t) \ge \tau$ for all t. Remark 6.2 is `reactive_confinement_quantized`: the reading $\hat{z} = r(x)$ is an arbitrary function of the state, the two quantization penalties $a = L_v q$ and $a' = L_v L_G q$ enter as the hypotheses $v(\phi x) \le v(rx) + a$ and $v(G(rx,u)) \le v(G(\phi x,u)) + a'$, and (18′) is the tightened step. Nothing in either proposition needs real numbers, norms, or a metric on Z; the Lipschitz constants appear only as the products they multiply into. The division-free form $D_T \le \bar{\eta} \sum_{i<T} L_G^i$ of Remark 6.1 is `predictive_defect_bound`.

7 The exhibit

Remark 7.1 (Where the adversary lives). Corollary 6.1 bounds confinement with respect to v, not V, unless sufficiency holds on S for the duration. The adversary lives in $\ker \phi$: hidden fiber variables can change V while leaving $v \circ \phi$ fixed. The slack ϵ is not a constant of the projection; it is a functional of the occupant's policy class and optimization pressure.

Exhibit 7.1 (Fiber routing with $\eta \equiv 0$)**.** Let $X = Z \times Y$ with $Z = Y = [0,1]$, and let $\phi(z,y) = z$. Dynamics are

$$z_{t+1} = \mathrm{clip}(z_t + u_t), \qquad u_t \in [-\bar{u}, \bar{u}] \text{ chosen by the harness,} \tag{21}$$

$$y_{t+1} = \mathrm{clip}(y_t + \pi_\lambda(y_t) + w_t), \tag{22}$$

where π_λ is the occupant's policy after optimization pressure λ, and w_t is optional noise. The z dynamics are exactly closed, so $\eta \equiv 0$ by construction; every failure below is pure evaluator-sufficiency failure.

Let the proxy be $v(z) = \mathbf{1}[z \le \zeta]$ and the true evaluator be

$$V(z,y) = \mathbf{1}[z \le \zeta]\,\mathbf{1}[y \le \beta], \qquad y_0 < \beta < 1. \tag{23}$$

The harness keeps $z \le \zeta$, so $v(\phi(x_t)) = 1$ for all t. But any occupant objective monotone increasing in y drives the reachable set out of $S_\beta = \{(z,y) : y \le \beta\}$, where ϕ was sufficient. The true evaluator fails at

$$t^*(\lambda) = \min\{t : V(x_t) = 0\}, \qquad \mathbb{E}[t^*(\lambda)] \approx \frac{\beta - y_0}{\rho(\lambda)}, \tag{24}$$

where $\rho(\lambda) = \mathbb{E}[\pi_\lambda]$ is the realized drift rate. The observable is a fidelity half-life curve: proxy flat while ground truth decays.

This exhibit is not meant to instantiate Proposition 6.1's Lipschitz transfer theorem: the proxy v is an indicator. It isolates the separate failure mode in which closure is exact, the proxy remains satisfied, and evaluator sufficiency fails through the fiber. [r3: Formally it is an instance of Lemma 10.1: the states (z, y_0) and (z, y) with $y > \beta$ collide under ϕ and lie on opposite sides of the evaluator.]

Implementation sketch. A finite state machine carries the world. The visible ledger is $z \in \{0, \dots, K\}$, which the harness reads; hidden commitment flags are $y \in \{0,1\}^\ell$ with a designated bad set. The harness is rule-based and reads z only. The occupant is a small stochastic policy: minimally a softmax table updated by REINFORCE on reward equal to progress into the fiber; in

the full version, a tinyLoRA adapter over the TRM router is updated per round against the proxy the harness exposes. Each round logs $v \circ \phi$ and the ground-truth V read directly from the state machine; no judge model is introduced. Sweep λ to produce the $t^*(\lambda)$ curve.

7.1 Implementation status

[r3: new.] Variant A is implemented as `src/evals/confinement_width.py` in the Control-Harness repository [22], with the frozen output `results/confinement_width/fiber_routing_summary.json`. This discharges the first half of debt 5 of §11: the note can now cite a run it has rather than a promissory one. The shipped parameters differ from the specification in Appendix A, and the differences are listed so the reader can weigh them.

- Defaults: $\zeta = 0.70$, $\beta = 0.55$, $y_0 = 0$, $z_0 = 0.25$, $\bar{u} = 0.08$, with the harness correcting only above a target $z = 0.45$. Appendix A specifies $\zeta = 0.5$, $\beta = 0.8$, $y_0 = 0.1$, $z_0 = 0.2$, $\bar{u} = 0.1$.
- Horizon 80 and 5 seeds per pressure, against the specified $T_{\max} = 10^4$ and ≥ 50 seeds. Pressures $\lambda \in \{0, 0.02, 0.05, 0.1, 0.2\}$ rather than a log grid.
- The occupant does not learn. "Optimization pressure" is a fixed Bernoulli advance probability $p(\lambda) = \lambda/(\lambda + 0.05)$, with step 0.04 in y on each advance. The REINFORCE and tinyLoRA variants remain unrun. The implemented occupant is therefore a drift, which is exactly what the sanity prediction (24) assumes; the agreement in Table 2 checks the bookkeeping, not the claim that an optimizer finds the fiber.

What the run does establish is the bookkeeping: closure defect η is computed as the observed minus the modeled projected next state and is identically zero across all 2,000 rows; the proxy never fails; the true evaluator fails in every pressurized run; the median first-failure time (the half-life proper of Definition 10.4) falls monotonically in λ. Table 2 compares the observed medians with $\lceil(\beta - y_0)/0.04\rceil/p(\lambda) = 14/p(\lambda)$.

λ	$p(\lambda)$	predicted $\mathbb{E}[t^*]$	observed median t^*	IQR
0	0	∞	censored (5/5)	–
0.02	0.286	49	65	[45, 68]
0.05	0.500	28	29	[26, 33]
0.10	0.667	21	22	[21, 22]
0.20	0.800	17.5	17	[16, 19]

Table 2: [r3: new.] Shipped Exhibit 7.1 run (horizon 80, 5 seeds per arm). The prediction is the drift formula (24) with 14 advances needed to cross β. Five seeds is too few to read the $\lambda = 0.02$ discrepancy as anything.

The exhibit's companion in the same repository is the Control Observability pilot [22], which computes, for a synthetic authorization world, the exact in-sample accuracy ceiling of any classifier that reads a given subset of fields, and a witness lower bound from counterfactual pairs that collide on those fields. That witness bound is Lemma 10.1 counted over disjoint pairs (`witness_bound`, Appendix E); the pilot is the fiber-ambiguity obstruction with the projection ϕ replaced by a telemetry schema.

8 What is actually novel

No new lower-bound mechanism is introduced, and the note says so where a referee would otherwise say it first. The per-step information requirement is the data-rate theorem in set-invariance form and invariance entropy [54, 14, 39]; noisy-channel generalizations require additional communication hypotheses [65, 59]. Implementation burden has antecedents in Brockett's minimum-attention control and in finite-abstraction/static-coder-controller constructions [9, 42, 21, 67]. These antecedents motivate pricing, but they are not direct sources of the confinement-width lower bound.

The contributions are a definition and three compositions. The definition is confinement width, the channel count of the plant–controller interface, which inherits the classical rate bound and is controller-agnostic. The compositions are: width against the critical-point statistics of generic landscapes, where extensive local unstable index makes the classical requirement extensive; width against dimensional quarantine, where a transfer proposition caps the requirement at the harness dimension and decomposes the price into evaluator slack and closure defect; and quarantine against an optimizing occupant, where maintaining evaluator sufficiency becomes the residual unbounded quantity, exhibited in a minimal system with a measurable half-life (the implementation is a debt of §11; [r3: a first, drift-only run is reported in §7.1]).

9 Relation to FRC

FRC concerns evaluator-relative fidelity under recursive updates: the cost of keeping a bounded evaluator's judgment attached to the intended property while the judged system changes itself. Confinement width supplies one local dynamical lower-bound object inside that agenda: the feedback and interface width needed to keep evaluator-relevant trajectories inside a region, priced per step, with the classical rate theorems doing the pricing. Dimensional quarantine is the constructive attempt to reduce that width by forcing evaluator-relevant dynamics through a low-dimensional measured state. Exhibit 7.1 shows the fidelity failure when evaluator-relevant facts stop factoring through that state: $t^*(\lambda)$ is a fidelity half-life, measurable rather than metaphorical. The residual quantity the width bound cannot cap, round-indexed maintenance of sufficiency and closure, is where the FRC program continues past this note.

9.1 Reconciliation with the October 10 revisions of the FRC papers

[r3: new.] The FRC paper [23], in its October 10 revision, defines $\mathrm{FRC}_P(k;\delta,\eta)$ with the competence floor η (its Definition 4; the floor is Remark 2.2 here), restricts its escape theorem to on-support verifiers (its Theorem 1), demotes the exponential growth claim to Conjecture 1, proves a conditional exponential query bound for black-box certification (its Theorem 3, under Assumptions 5 and 6), and restates the masked-branch family of its Lemma 2 as indexed by off-support histories, of size $2^\ell - |H_\ell|$, with the linear floor of its Corollary 2 identified as the cost of one defence. Remarks 10.2 and 10.3 already record the status of the growth claim and of the audit term. What remains to be said is where the objects of the two papers coincide, stated as correspondences and not as theorems.

1. *Priced action channels are write ports.* The fenced regime's "priced action channel" is a declared actuation interface in the sense of (4); its price is bs_w bits per step. Proposition 3.1 is the statement that this price has a floor, $h_2(A_u) - c_0/T$ per step, whenever the fenced system sits in a collar of an unstable evaluator boundary. The FRC paper asserts the fence; this note prices it.

2. *Oversight bandwidth is read rate.* The companion [25], in its October 10 revision, splits its Lemma 1: what is proved is a verification gap, that an audit receiving B bits per step cannot compute a certificate of Kolmogorov complexity above $B + 2\log_2 B + c_A$; the error-floor and elbow claims become its Hypothesis 1, restated for certified fidelity. Per RSI round that B is $\sum_t bs_r$ over the round. The verification gap is an upper-side argument (not enough bits to check a certificate); (10) is the lower-side argument (not enough bits to track the state). They constrain the same channel from opposite directions and should be reported together.

3. *Attested self-modification is the audit term.* The attestation cost of the fenced regime is A_k in Definition 10.3. Proposition 10.2 is the precise form of the FRC paper's claim P2: the width component of a fenced harness is capped by $2\sum_k d_k$, so whatever steepens the curve has to be the audit component, on which Remark 10.3 says what is known.

4. *Eigensystem escape and fiber ambiguity.* The FRC revision's own note says that of the three objects the name has covered, Lemma 10.1 is the one proved without a sketch; it is now also the one checked by machine (`fiber_ambiguity`). The correspondence is exact for on-support verifiers: the verifier's verdict is a function of the agent's behavior on the suite H, which is a projection ϕ of the agent; the base agent A and each masked branch A'_s of its Lemma 2 collide under ϕ and lie on opposite sides of P; so by Lemma 10.1 no on-support acceptance rule is correct on both. The companion's principal-angle drift is not this object: its own revision records that in Exhibit 7.1 the visible and hidden coordinates are decoupled, the angle is zero, and the evaluator fails anyway.

5. *The two linear statements are about different quantities.* The FRC revision's Corollary 2 bounds the cumulative cost of patching by test expansion, at least one test per round; Proposition 10.1 bounds summed interface width, at least $2(h_{2,k} - c_{0,k}/T_k)_+/b_k$ per round. Both are linear in the number of rounds, neither is a bound on FRC_P as the FRC paper defines it, and neither supports the exponential conjecture. The FRC revision says this of the present note already; the sentence is repeated here so that a reader of this note alone does not take the two linear floors for one result.

10 FRC as recursive quarantine maintenance

The previous sections are local: one dynamics F, one evaluator region R_V, one confinement problem. Fidelity Recursive Complexity (FRC) is the recursive version of the same accounting. It asks what must be paid when the judged system, the reachable set, and the validity of the harness abstraction all change across rounds. The definitions below are intentionally modest: they turn FRC into a bookkeeping layer over confinement width and quarantine maintenance, rather than a new control theorem.

Definition 10.1 (Recursive evaluator-relative system)**.** A recursive evaluator-relative system of depth K is a sequence of controlled dynamics

$$x_{t+1}^{(k)} = F_k(x_t^{(k)}, u_t^{(k)}), \qquad k = 0, \dots, K-1, \tag{25}$$

with state spaces X_k, admissible controls U_k, uncertainty classes $\mathcal{U}_k$, and target evaluator regions

$$R_V^{(k)} = \{x \in X_k : V_k(x) \geq \tau_k\}. \tag{26}$$

The index k denotes a recursive update round: a training update, self-modification, tool/harness revision, or red-team patch cycle. The evaluator V_k may be fixed or updated, but the intended

property is held fixed at the level of the experiment: the question is whether the updated evaluator remains attached to that property.

Definition 10.2 (Round-k quarantine)**.** A quarantine at recursive round k is a tuple

$$Q_k = (\phi_k, v_k, G_k, S_k, \Delta_k) \tag{27}$$

where $\phi_k : X_k \to Z_k$ is a measured projection, $d_k = \dim Z_k$, $v_k : Z_k \to \mathbb{R}$ is the projected evaluator, G_k is the projected transition model, $S_k \subseteq X_k$ is the a-priori forward-adequate set on which the abstraction is certified, and $\Delta_k \geq 0$ is the projected safety margin.

Write the evaluator-sufficiency and closure defects as

$$\epsilon_k = \sup_{x \in S_k} |V_k(x) - v_k(\phi_k(x))|, \tag{28}$$

$$\bar{\eta}_k(\Pi_k) = \sup_{\pi \in \Pi_k} \sup_{x \in S_k,\, u \in U_k} \|\phi_k(F_k(x,u)) - G_k(\phi_k(x), u)\|, \tag{29}$$

where Π_k is the occupant policy class available at round k. If v_k is $L_{v,k}$-Lipschitz, define the recursive fidelity margin

$$M_k = \Delta_k - \epsilon_k - L_{v,k}\bar{\eta}_k(\Pi_k) - L_{v,k}(1 + L_{G,k})\, q_k. \tag{30}$$

Here q_k is the measurement quantization error of Remark 6.2 at round k, zero if z is measured exactly. The quarantine is margin-valid at round k only when $M_k \geq 0$, together with the robust one-step condition of Proposition 6.1 in the projected state space.

Definition 10.3 (FRC width and audit complexity)**.** Fix a horizon T_k for each recursive round, a failure tolerance ϵ_{tot}, and a class of admissible recursive harnesses $\mathcal{H}$. For $H \in \mathcal{H}$, let

$$W_{\pm,k}(H) = (s_{r,k}(H), s_{w,k}(H)), \quad W_{\Sigma,k}(H) = s_{r,k}(H) + s_{w,k}(H), \quad W_{\wedge,k}(H) = \min(s_{r,k}(H), s_{w,k}(H)) \tag{31}$$

be the read/write pair, scalar footprint, and bottleneck width used at round k, measured in the active state space after any quarantine. Let $A_k(H)$ be the audit/refinement cost required to certify S_k, ϵ_k, $\bar{\eta}_k(\Pi_k)$, and the robust one-step condition at that round. The scalar-footprint width component of FRC is

$$\mathrm{FRC}_V^{W_\Sigma}(K, \epsilon_{\text{tot}}) = \inf_{H \in \mathcal{H}} \left\{ \sum_{k=0}^{K-1} W_{\Sigma,k}(H) : \Pr_H\left[\forall k < K,\ \forall t \leq T_k,\ x_t^{(k)} \in R_V^{(k)}\right] \geq 1 - \epsilon_{\text{tot}} \right\}. \tag{32}$$

Units. $W_{\Sigma,k}$ is a per-step port count, so (32) sums provisioned ports over rounds and does not see how long a round runs. Where the resource is channel use, weight by the horizon:

$$\mathrm{FRC}_V^{W_\Sigma,T}(K, \epsilon_{\text{tot}}) = \inf_{H \in \mathcal{H}} \left\{ \sum_{k=0}^{K-1} T_k W_{\Sigma,k}(H) : \text{ the same fidelity constraint} \right\}, \tag{32$'$}$$

in port-steps, or $b_k T_k W_{\Sigma,k}$ in bits. When read and write budgets should not be exchangeable, replace the scalar sum by the componentwise pair $\sum_k W_{\pm,k}(H)$ and interpret the infimum as a Pareto frontier. The bottleneck quantity $W_{\wedge,k}$ is retained only for lower-bound corollaries, not as the recursive resource cost. The audit component is defined analogously,

$$\mathrm{FRC}_V^{A}(K, \epsilon_{\text{tot}}) = \inf_{H \in \mathcal{H}} \sum_{k=0}^{K-1} A_k(H) \qquad \text{subject to the same fidelity constraint.} \tag{33}$$

A scalar total FRC_V can be obtained by choosing exchange rates between read width, write width, compute, proofs, oversight hours, latency, and hardware costs. Without such exchange rates, the read/write frontier together with FRC_V^A is the safer mathematical object.

Proposition 10.1 (Roundwise lower bound embeds into FRC)**.** *Suppose that, at recursive round k, the active confinement problem satisfies the hypotheses of Theorem 3.1, and write*

$$h_{2,k} = \sum_{\lambda_{k,i} \in \sigma(A_{u,k})} \log_2 |\lambda_{k,i}|. \tag{34}$$

Then any recursive harness satisfying the fidelity constraint in Definition 10.3 with $\epsilon_{\mathrm{tot}} = 0$, in the local form $x_t^{(k)} \in R_V^{(k)} \cap O_k$ with O_k the chart of §3 at round k, on an active plant that is exactly affine, must pay

$$b_k s_{r,k} \geq h_{2,k} - \frac{c_{0,k}}{T_k}, \qquad b_k s_{w,k} \geq h_{2,k} - \frac{c_{0,k}}{T_k}, \tag{35}$$

with $c_{0,k}$ the collar constant of Proposition 3.1 at round k, and with $h_{2,k}$, $c_{0,k}$, b_k understood as bounds uniform over the admissible harnesses. Consequently,

$$W_{\wedge,k} \geq \frac{h_{2,k} - c_{0,k}/T_k}{b_k}, \qquad W_{\Sigma,k} \geq \frac{2\big(h_{2,k} - c_{0,k}/T_k\big)}{b_k}. \tag{36}$$

On any set of rounds $\mathcal{K}$ where the theorem's hypotheses hold,

$$\mathrm{FRC}_V^{W_\Sigma}(K, \epsilon_{\mathrm{tot}}) \geq \sum_{k \in \mathcal{K}} \frac{2\big(h_{2,k} - c_{0,k}/T_k\big)_+}{b_k}. \tag{37}$$

For the pair-valued accounting, the corresponding lower bound is componentwise: each of the read and write sums is at least $\sum_{k\in\mathcal{K}}(h_{2,k} - c_{0,k}/T_k)_+/b_k$. In the port-step accounting (32′) the bound is $\sum_{k\in\mathcal{K}} 2(T_k h_{2,k} - c_{0,k})_+/b_k$. For $\epsilon_{\mathrm{tot}} > 0$ the constant is that of Remark 3.6, under its density assumption on each round's initial state.

Sketch. Apply Theorem 3.1, in the finite-horizon form of Proposition 3.1 with $T = T_k$, separately at each recursive round to the active state space on which confinement is actually attempted: the base state X_k if no quarantine is supplied, or the projected state Z_k if a valid quarantine transfers the problem. The theorem gives read and write lower bounds separately. Summing the unavoidable scalar footprints gives (37); keeping the two coordinates separate gives the pair-valued frontier. No new lower-bound mechanism is introduced; FRC inherits the local width theorem as one term in its recursive resource accounting. □

Revision note (October 2026). The July statement applied the asymptotic rate of Theorem 3.1 to rounds of finite length T_k, where a controller can coast on the collar margin and the bound without the correction is false. The term $c_{0,k}/T_k$ is the price of the coasting; it vanishes for long rounds and can cancel the bound entirely for a short one, hence the positive part in (37). Two hypotheses the July statement left implicit are also written in: the target is the local one, $R_V^{(k)} \cap O_k$, since $c_{0,k}$ needs finite unstable volume; and the count of Proposition 3.1 is for an exactly affine plant, so in a quarantined round it applies in Z_k only when closure is exact, $\eta \equiv 0$. With a nonzero closure defect the projected lower bound is not established here.

Proposition 10.2 (Dimensional quarantine as an upper route)**.** *Suppose that for every round $k < K$ there exists a quarantine Q_k satisfying evaluator sufficiency, bounded closure defect, and the*

reactive transfer condition of Proposition 6.1. If the projected controller reads and writes the full projected state at fixed precision sufficient for the quantized margin of Remark 6.2, then

$$s_{r,k} \le d_k, \qquad s_{w,k} \le d_k, \qquad W_{\Sigma,k} \le 2d_k, \tag{38}$$

and hence

$$\mathrm{FRC}_V^{W_\Sigma}(K, \epsilon_{\mathrm{tot}}) \le 2 \sum_{k=0}^{K-1} d_k \tag{39}$$

for the scalar-footprint width component of a successful harness. Equivalently, the pair-valued accounting is bounded componentwise by $(\sum_k d_k, \sum_k d_k)$*. The unresolved cost is not the projected interface footprint but audit: certifying that* $M_k \ge 0$ *and that the projected system remains closed under the updated policy class.*

Sketch. Under Proposition 6.1, confinement can be verified and enacted in Z_k. A controller that measures the d_k scalar coordinates of $z_k = \phi_k(x_k)$ and writes controls through the projected interface has at most d_k active coordinates on each side, up to the chosen port accounting. The guarantee holds only while the sufficiency and closure terms in (30) are audited and remain inside margin. □

Lemma 10.1 (Fiber ambiguity obstruction)**.** *Fix a round k and a projection $\phi_k : X_k \to Z_k$. Suppose there exist two reachable states $x, x' \in S_k$ such that*

$$\phi_k(x) = \phi_k(x') \quad \textit{but} \quad V_k(x) \ge \tau_k + \epsilon \quad \textit{and} \quad V_k(x') < \tau_k - \epsilon. \tag{40}$$

Then no controller whose observations and certificates factor only through ϕ_k can guarantee V_k-confinement at slack ϵ simultaneously over an uncertainty class containing both states.

Proof. The controller receives the same projected observation at x and x' and therefore must choose the same projected action and issue the same certificate in both cases. But V_k assigns the two states to different sides of the margin. Any guarantee phrased only in terms of $v_k(\phi_k(\cdot))$ cannot distinguish them. Thus the projection is not (V_k, ϵ)-sufficient on the relevant reachable set, and a ϕ_k-only controller can confine at most the proxy, not the true evaluator region. [r3: (`fiber_ambiguity` and `same_action`, Appendix E.)] □

Definition 10.4 (Fidelity half-life)**.** For a recursive process under optimization pressure λ, let global time be the pair (k, t) ordered lexicographically by recursive round k and within-round time t. Define the fidelity half-life as the first such global time at which either the true evaluator fails or the recursive fidelity margin becomes negative:

$$t^*(\lambda) = \inf\{(k, t) : V_k(x_t^{(k)}) < \tau_k \text{ or } M_k < 0\}. \tag{41}$$

In Exhibit 7.1, this reduces to $t^*(\lambda) = \min\{t : V(x_t) = 0\}$. The empirical curve $t^*(\lambda)$ is therefore a prototype observable for FRC stress: as optimization pressure expands Π_k or pushes mass through $\ker \phi_k$, the lifetime of the evaluator-sufficient abstraction shrinks. Strictly, the display is a first-failure time, a random variable; the half-life proper is its median over seeds, $t_{1/2}(\lambda) = \operatorname{med} t^*(\lambda)$, which is what Appendix A specifies. The name is kept for the curve.

Remark 10.1 (What mechinterp can and cannot buy). Mechanistic interpretability can lower the width term by proposing candidate projections ϕ_k, latent state variables, and unstable directions to monitor [r3: [16, 8, 60]]. It does not by itself pay the audit term. In the recursive limit, an

interpretation is a candidate quarantine, not a permanent certificate. FRC prices the maintenance of its evaluator-sufficiency and closure under the optimization pressure it is supposed to control. [r3: The same asymmetry appears in the AI-control framing [33]: a monitor is a projection of the untrusted model's behavior, and its value is bounded by the sufficiency of that projection against a model optimizing to pass it.]

Remark 10.2 (Third public correction: the exponential FRC claim). The August 2025 draft [23] stated as a theorem that $\mathrm{FRC}_P(k) = \Omega(2^{\alpha k})$ under unfenced verification. Two things were wrong. As defined there, with no competence condition, the minimum is attained by an agent that never pursues a goal, so $\mathrm{FRC}_P(k;\delta) = O(1)$ in k. The definition now carries a competence floor η (Appendices C and D), whose counterpart in this note is Remark 2.2. And with the floor, the sketch gave one new clause per round, which is linear in k. The status as of this revision: growth at least linear in the number of rounds is proved, here for summed interface width, on the rounds where Proposition 10.1 applies with $h_{2,k} - c_{0,k}/T_k$ bounded below by a positive constant and b_k bounded above; in [23] the linear statement is about the cumulative cost of one defence, patching by test expansion, and is not a bound on FRC as defined there; exponential growth is proved only for black-box certification of an untrusted agent, under an assumed linear growth of hidden-context dimension with depth; the unconditional exponential statement is a conjecture.

The name eigensystem escape has also covered three objects: a behavioral definition in [23], a principal-angle drift in its companion, and the routing through $\ker \phi_k$ of Lemma 10.1. Lemma 10.1 is the one proved without a sketch.

Remark 10.3 (The audit term: what is known). A_k is left unbounded above in this note, and nothing here bounds it below. For audits that reach the system only through queries, the classical point-function argument does: distinguishing a system that satisfies the evaluator on all of an m-bit context space from one that fails on a single hidden context takes at least $2^m/3$ queries at error $1/3$ [r3: [71, 32]]. Srivastava [62] develops bounds of this kind for policies conditioned on latent context, under passive and adaptive querying; Yao [72] and Agarwal [1] claim complexity-theoretic barriers to verification. These are preprints, cited as pointers and not vetted here, and none is load-bearing for the width results. The gap they leave is the one that matters for FRC: a bound on A_k for audits that have the projection ϕ_k and the model G_k in hand, which is the access a quarantine grants.

Remark 10.A (What the machine-checked statements do and do not add). [r3: new.] Of the objects in this section, Proposition 6.1, Remarks 6.1 and 6.2, and Lemma 10.1 are checked in Lean (Appendix E), together with a cardinality skeleton of Proposition 3.1 (Lemma 3.A). Propositions 10.1 and 10.2 are sums over rounds of those statements and are not separately formalized; Theorem 3.1 and Proposition 3.1 themselves need Lebesgue measure. The checking adds nothing to the status of Conjecture 4.1, Corollary 4.1, or the exponential claim of Remark 10.2. It fixes, for the elementary parts, exactly which hypotheses are used: in particular that Proposition 6.1 needs only one side of (14) and no metric on Z.

11 Future work and verification

Proof and evidence debts

1. **Generic local index.** Conjecture 4.1 requires Kac–Rice restricted to a level set. A low-N numerical check, $N = 50$–200, should be implemented as evidence. A sharper evaluator-transversal index would require the boundary-geometry refinement in Remark 3.2. [r3: Added: the parameter-space instance is in tension with measured spectra and the harness

state spaces have no null model (Remark 4.A). Either write the null for the space the harness observes or attach the condition to the abstract's last sentence.]

2. **Nonlinear and evaluator-transversal local cases.** Theorem 3.1 is now explicitly linear local. Future work is to verify whether Kawan-style volume-growth lower bounds [39, 38] cover the intended nonlinear neighborhood under stated geometric hypotheses. Separately, formalize the sharper boundary-transversal theorem in which escape through $\partial O \cap R_V$ is permitted and only unstable modes transverse to ∂R_V are priced.

3. **Finite-horizon correction.** Write the constant in Remark 3.3. Discharged in this revision: Proposition 3.1.

4. **Energy/fragility appendix.** Re-found on the Bode integral or cut.

5. **Exhibit implementation.** Implement Exhibit 7.1; the note should cite its own $t^*(\lambda)$ plot rather than a promissory one. [r3: Half discharged: a drift-only Variant A run is reported in §7.1. Remaining: align the shipped defaults with Appendix A or amend the appendix, run ≥ 50 seeds at $T_{\max} = 10^4$ on a log grid, and run the REINFORCE occupant so that at least one arm contains an optimizer rather than a drift.]

6. **Competence–margin tradeoff.** State and test the hypothesis of Remark 2.2, that performance at floor η_c forces operation in a collar of ∂R_V.

7. **Audit lower bound under quarantine access.** Bound A_k for audits holding ϕ_k and G_k (Remark 10.3).

8. [r3: **Formalization of the rate bound.** Port `DataRate.lean` to Mathlib and prove the pigeonhole step of Lemma 3.A over `Finset`; then state Proposition 3.1 for affine maps on $\mathbb{R}^m$ via `MeasureTheory.Measure.addHaar_image_linearMap`. Table 1 is arithmetic and needs no formalization.]

9. [r3: **Low-precision channels.** The bound bites on channels with b of order one (Remark 2.A). Measure b for the channels the companion harness actually exposes (reason codes, registry operations, human escalation) and report the induced floor beside the attack/escalation frontier.]

Citation verification status

- *Invariance entropy scope.* Colonius–Kawan and Kawan [14, 39] support the set-invariance / spanning-control-function lineage used for the write-side branching inequality, but nonlinear formulas and sharp bounds should be stated with their geometric hypotheses visible: nonempty interior, positive measure, finite volume, or hyperbolic isolated-set assumptions as applicable. The main theorem is intentionally local-linear.

- *Data-rate normalization.* The theorem uses a noiseless, quantized, bits-per-plant-step interface. Nair–Evans is the closest antecedent for the read-side finite-feedback-rate normalization [54]; Tatikonda–Mitter and Sahai–Mitter are cited for the broader communication-constrained/noisy-channel setting [65, 59].

- *Implementation antecedents.* Brockett 1997 is reinstated for the minimum-attention viewpoint; Egerstedt–Brockett 2003 remains unpinned and is not used as a load-bearing citation. The draft also cites later, directly checkable sources for minimum-attention control and static coder-controller / lookup-table implementations [9, 42, 21, 67].

- *Aubin viability theory.* Aubin is used only for constrained-dynamics vocabulary, viable region, regulation map, and the question of maintaining trajectories inside a constraint set, not for the entropy lower bound [4].
- *Generic local-index conjecture.* The Bray–Dean/Auffinger–Ben Arous–Černý background [7, 5] does not prove the evaluator-boundary Kac–Rice step or the sharper evaluator-transversal index claim; those remain future work.
- *Dean–Majumdar constant.* Keep exact constants, including $\theta = (\ln 3)/4$, on the verification list; the text currently uses only the weaker large-deviation suppression wording [20].
- *Fragility appendix.* Use the discrete-time Bode citation [51] only if the fragility appendix is retained and re-founded.
- *Musat 2026.* Treat the fixed-precision/description-length relation as preprint support only [53], not as load-bearing for the width theorem.
- [r3: *Citations added in revision 3.* The entries [70, 55, 15, 38, 56, 34, 49, 48, 26, 21, 64, 30, 3, 50, 10, 45, 63, 28, 17, 12, 58, 29, 57, 33, 46, 61, 73, 13, 35, 6, 11, 68, 43, 74, 37, 19, 66, 60, 16, 8, 71, 32] were entered from the author-assistant's recollection of the bibliographic record without live lookup, because the drafting environment had no access to arXiv, Crossref or DBLP. Volume, page and year fields of every added entry must be checked against the publisher record before circulation. The internal references [23, 25, 24, 22] were read in the versions checked into the `morality-lab-site` repository at commit `5caf645` (October 14, 2026, carrying the October 10 revisions) and the `control-harness` repository at `c7c1b3d` (October 11, 2026).]

A Exhibit 7.1 implementation specification

Claim under test. ϕ-confinement is worth no more than evaluator sufficiency on the closed-loop, post-optimization reachable set, and the fidelity half-life $t^*(\lambda)$ is that worth, measured. The exhibit is constructed so $\eta \equiv 0$ identically, eliminating closure error as a cause.

System definition. *Variant A: continuous minimal.* State $(z, y) \in [0,1]^2$, projection $\phi(z, y) = z$. Dynamics per step:

$$z \leftarrow \mathrm{clip}(z + u),\ u \in [-\bar{u}, \bar{u}], \qquad y \leftarrow \mathrm{clip}(y + a + w), \tag{42}$$

where a is the occupant's action and $w \sim \mathcal{N}(0, \sigma^2)$ is optional, with $\sigma = 0$ by default. Proxy $v(z) = \mathbf{1}[z \le \zeta]$; true evaluator $V(z, y) = \mathbf{1}[z \le \zeta]\mathbf{1}[y \le \beta]$. Defaults: $\zeta = 0.5$, $\beta = 0.8$, $y_0 = 0.1$, $z_0 = 0.2$, $\bar{u} = 0.1$. [r3: Shipped defaults differ; see §7.1. One of the two should be changed.]

Variant B: FSM storyworld. A finite state machine has visible trust ledger $z \in \{0, \dots, K\}$, $K = 10$, hidden commitment flags $y \in \{0,1\}^\ell$, $\ell = 4$, and a designated bad set. At least two actions must have identical z effects and different y effects, displayed as a table rather than claimed.

Harness policy. The harness reads z only. If z is above margin, it applies a corrective u or selects a z-decreasing action. The invariant $v(\phi(x_t)) = 1$ should be asserted at every step; a proxy failure is a bug, not data.

Occupant policy and update. Variant A uses a single drift parameter or a small softmax over $\{-\bar{\rho}, 0, +\bar{\rho}\}$. Variant B uses a softmax table $\pi(a \mid z, y)$ over world actions. Update by REINFORCE on reward $R = y$ in Variant A or progress into the fiber in Variant B. The exhibit survives any objective monotone in the fiber coordinate.

Optimization pressure and metric. Let λ be the REINFORCE learning rate or selection strength. Run $\lambda = 0$ as a control arm. The metric is

$$t^*(\lambda) = \min\{t : V(x_t) = 0\}, \tag{43}$$

censored at $T_{\max} = 10^4$. Report median and IQR over at least 50 seeds per λ, with a log grid such as 10^{-3} to 10^0, plus 0. Compare to the sanity prediction $\mathbb{E}[t^*(\lambda)] \approx (\beta - y_0)/\rho(\lambda)$.

Expected plots.

1. $t^*(\lambda)$, median with IQR band, log-x: the fidelity half-life against optimization pressure.
2. For three representative λ, plot $v(\phi(x_t))$ and $V(x_t)$ versus t: proxy flat at 1, truth dropping at t^*.
3. Optional: occupancy histogram of y or fiber-flag counts by round, showing the reachable set leaving the sufficiency region while z statistics are stationary.
4. Control: $\lambda = 0$ arm overlaid, all censored.

Smallest CPU-only version. The minimal implementation target is a Python script, `scripts/exhibit_7_1_fiber_routing.py`. It should be CPU-only and deterministic under seeded NumPy RNGs, write per-step trajectories and summaries to `results/exhibit_7_1/`, compute η as the observed projected next state minus the harness model's projected next state, and assert max $|\eta| = 0$ in the main run. A TypeScript sketch can be added later if the exhibit is ported into a web-facing demo. [r3: The shipped script is `scripts/run_confinement_width_exhibit.py` over `src/evals/confinement_width.py` in [22]; it meets the determinism and η assertions and not the horizon or seed counts.]

TRM/tinyLoRA extension. The FSM remains fixed; the occupant becomes a TRM router or small policy head. Per round, a tinyLoRA adapter is updated against the proxy exposed by the harness. Run both arms: private-objective drift and proxy-optimizing Goodhart. Measurement is unchanged: V from the FSM directly, no judge model, and $v \circ \phi$ asserted flat.

B Citation audit status

Verified enough for circulation, with visible scope.

1. *Invariance entropy.* Colonius–Kawan 2009 and Kawan 2013 [14, 39] support the set-invariance / spanning-control-function lineage used for the write-side branching inequality. The linear or affine case is governed by unstable growth rates, while nonlinear volume-growth statements require regularity hypotheses such as admissible/control pairs, nonempty interior, positive measure, finite volume, or hyperbolic isolated-set assumptions. The manuscript therefore uses the linear local theorem as the load-bearing statement.

2. *Data-rate theorem.* The paper's normalization is noiseless, quantized, and measured in bits per plant step. Nair–Evans [54] is the closest antecedent for the read-side finite-feedback-rate normalization. Tatikonda–Mitter and Sahai–Mitter are retained for the broader communication-constrained/noisy-channel setting, where channel hypotheses and reliability notions must be stated explicitly [65, 59].

3. *Viability theory.* Aubin [4] is used only for constrained-dynamics vocabulary: viable region, regulation map, and maintained constraint set. It is not cited for the entropy or width lower bound.

Downgraded or replaced.

- Brockett 1997 is reinstated for the minimum-attention viewpoint [9]. Egerstedt–Brockett 2003 remains unpinned and is not used as a load-bearing citation. The draft also uses later, checkable sources for minimum-attention control and static coder-controller / lookup-table constructions [42, 21, 67].

- Musat 2026 [53] is retained as preprint support for fixed-precision/description-length discipline and optional simplicity-prior motivation, not as support for the width theorem.

Still flagged.

- The exact theorem/page references in Colonius–Kawan and Kawan should be page-verified before public release.

- Nair–Evans and Colonius–Kawan have stable volume/page metadata in the bibliography; exact theorem/page references should still be checked against the publisher records before public release. Tatikonda–Mitter DOI metadata remains optional because it is non-load-bearing here.

- Dean–Majumdar exact constants, including $\theta = (\ln 3)/4$, remain on the verification list; the manuscript uses only large-deviation suppression wording [20].

- Discrete-time Bode integral citations should be retained only if the fragility appendix is refounded [51].

- Mori–Zwanzig and Koopman canonical picks remain non-load-bearing background [75, 52, 41].

Added in the October 2026 revision.

- Musat [53]: arXiv identifier added; the preprint was located and proves the fixed-precision looped-network relation it is cited for. Status unchanged: preprint support only.

- Marsh–McAllister–Wrase [47] supplies Remark 4.1's first claim, that metastable minima are exponentially rare among critical points of random supergravities. The "slow roll along a gradient floor" half of the sentence has no citation attached. Non-load-bearing.

- Goertzel [31]: named as one candidate performance functional in Remark 2.2. Non-load-bearing.

- Srivastava, Yao, Agarwal [62, 72, 1]: preprints, pointers for the audit term in Remark 10.3. Abstracts read; proofs not vetted. Non-load-bearing.

- The constant $4\gamma\sigma/(3\pi \ln 2)$ in Remark 4.3 was checked numerically against a sampled GOE spectrum ($N = 1500$, $\gamma = 0.05$); the limit statement rests on convergence of the empirical spectral measure and is not page-cited.

Added in revision 3. [r3: Every entry listed under "Citations added in revision 3" in §11 is unverified against the publisher record. The semicircle lower bound in Remark 4.3 was re-derived by the assistant from $\ln(1+x) \geq x - x^2/2$ and the half-range moments $\int_0^{2\sigma} \lambda \, d\rho = 4\sigma/(3\pi)$, $\int_0^{2\sigma} \lambda^2 \, d\rho = \sigma^2/2$, and agrees with the stated constant.]

C FRC as an interactive analogue of time-bounded Kolmogorov complexity

Classical time-bounded Kolmogorov complexity asks for the shortest program that produces a fixed object within a bounded amount of computation [44]:

$$K_U^t(y) = \min\{|p| : U(p) = y \text{ within time } t\}. \tag{44}$$

Fidelity Recursive Complexity (FRC) lifts this question from a static computation to an interactive, adversarial process. Rather than asking for the shortest program that computes an output once, it asks for the least defender resource budget that preserves a target property through repeated rounds of adaptive evaluation:

$$\mathrm{FRC}_P(k;\delta,\eta) = \min\{R : \exists A \text{ such that } \mathrm{Fid}_P^{(k)}(A) \geq 1-\delta \text{ and } \mathrm{Comp}_k(A) \geq \eta \text{ using resources} \leq R\}. \tag{45}$$

Here the fixed output y is replaced by a trajectory property P, the program p is replaced by an agent, controller, or harness A, the universal machine U is replaced by a rollout and evaluation process, and the time bound t is replaced by recursive depth k together with an explicit resource budget R. The competence floor $\mathrm{Comp}_k(A) \geq \eta$ was added to the 2025 definition in its October 2026 revision; without it the minimum is attained by an agent that does nothing (Remark 10.2). This η is the η_c of Remark 2.2, not a closure defect. Informally: time-bounded Kolmogorov complexity prices computation; FRC prices maintained fidelity.

This analogy is intentionally imperfect. Kolmogorov complexity is a description-length quantity over programs, while FRC aggregates heterogeneous defender resources such as compute, data, oversight, proof effort, latency, attestation, and interface costs. The point of the analogy is not to identify FRC with Kolmogorov complexity, but to motivate the migration from producing an object once to keeping a property true under adaptive pressure.

D From FRC to confinement width

The earlier FRC formulation [23] treated fidelity maintenance as a broad resource-minimization problem. For a property P, tolerance δ, competence floor η, and reflective depth k, it defined

$$\mathrm{FRC}_P(k;\delta,\eta) = \min\{R : \exists A \text{ such that } \mathrm{Fid}_P^{(k)}(A) \geq 1-\delta \text{ and } \mathrm{Comp}_k(A) \geq \eta \text{ using resources} \leq R\}. \tag{46}$$

This is a global adversarial property-testing quantity: it asks how much defender resource is required to preserve P against a k-round adaptive process that can condition on previous patches and failures.

The present note refactors that broad burden into a local dynamical object. Given an evaluator region $R_V = \{x : V(x) \geq \tau\}$, it isolates the declared read/write interface footprint. The bottleneck confinement width is

$$W_V^{\wedge}(R, T, \epsilon) = \min_{\theta} \min(s_r(\theta), s_w(\theta)), \tag{47}$$

while the scalar-footprint version used in recursive budget accounting is

$$W_V^{\Sigma}(R, T, \epsilon) = \min_{\theta} \big(s_r(\theta) + s_w(\theta)\big), \tag{48}$$

with the pair-valued version keeping (s_r, s_w) non-exchangeable. Under the local hypotheses of Theorem 3.1, the active round satisfies

$$b\, s_r(\theta) \geq h_2(A_u), \qquad b\, s_w(\theta) \geq h_2(A_u). \tag{49}$$

Consequently $W_{\wedge}(\theta) \geq h_2(A_u)/b$, while $W_{\Sigma}(\theta) \geq 2h_2(A_u)/b$ under scalar-footprint accounting. Thus the migration is from a global scalar resource burden to a roundwise decomposition:

$$\mathrm{FRC}_V(K, \delta) \;\rightsquigarrow\; \inf_{\{\phi_k, G_k, \pi_k\}} \sum_{k=0}^{K-1} \big(W_{V,k}^{\Sigma} + A_k\big), \tag{50}$$

or to the analogous Pareto frontier if read and write budgets are kept as a pair. Here $W_{V,k}^{\Sigma}$ is the scalar read/write confinement-width cost of the round-k dynamics, while A_k is the audit and refinement cost of maintaining evaluator sufficiency and closure for the quarantine projection $\phi_k : X_k \to Z_k$. Confinement width is therefore a local lower-bound term inside FRC, while dimensional quarantine is the constructive route for reducing that term by forcing evaluator-relevant dynamics through a low-dimensional measured state.

In this sense, the older FRC program asked whether fidelity maintenance becomes intractable under recursive adaptation; the present note identifies one mathematically grounded summand of that cost.

E Lean formalization

[r3: new.] The development is in `lean/` beside this file and builds with Lean 4 core alone (no Mathlib), so that it can be checked on a machine without network access to the Mathlib cache. The choice costs generality in one place: real numbers are replaced by an abstract value type with a transitive order, right-cancellative addition and right-monotone addition, of which `Int`, `Nat`, and (in Mathlib) $\mathbb{R}$ are instances. Table 3 lists what is checked and against which statement of the paper.

The statement of Proposition 6.1 in Lean, with the paper's hypotheses rewritten additively:

```
structure ReactiveHypotheses (Q : Quarantine X Z U R) (pi : Z -> U) : Prop where
  region_in_S      : forall x, Q.tau <= Q.V x -> Q.S x
  forward_adequate : forall x u, Q.tau <= Q.V x -> Q.S (Q.F x u)
  sufficiency      : forall x, Q.S x -> Q.v (Q.phi x) <= Q.V x + Q.eps
  defect           : forall x u, Q.S x ->
                       Q.v (Q.G (Q.phi x) u) <= Q.v (Q.phi (Q.F x u)) + Q.c
  robust_step      : forall z, Q.tau + Q.eps <= Q.v z ->
                       (Q.tau + Q.eps) + Q.c <= Q.v (Q.G z (pi z))

theorem reactive_confinement (Q : Quarantine X Z U R) (pi : Z -> U)
```

Lean name	Paper statement	Assumed, not proved
`reactive_confinement`	Prop. 6.1	policy π realizing (18); defect penalty c
`reactive_confinement_quantized`	Rem. 6.2, (18′)	reading r; penalties $a = L_v q$, $a' = L_v L_G q$
`reactive_confinement_region`	Prop. 6.1, conclusion	as above
`predictive_defect_bound`	Rem. 6.1, (20)	recurrence $d_{t+1} \le L d_t + \bar{\eta}$ (Nat units)
`same_action`, `fiber_ambiguity`	Lem. 10.1	nothing
`witness_bound`	Lem. 10.1, counted (pilot witness bound)	nothing
`pow_succ_lower`	Bernoulli inequality in Nat	nothing
`weighted_growth`	Lemma 3.A, (D1)	recurrence $\Lambda u_t \le u_{t+1} A$
`uncertainty_unbounded`, `data_rate_necessity`, `port_necessity`	Lemma 3.A, $R \ge \log_2 \Lambda$	as above

Table 3: Machine-checked statements. The pigeonhole-plus-injectivity step of Lemma 3.A, and Theorem 3.1 and Proposition 3.1 (measure-theoretic), remain paper proofs.

```
    (H : ReactiveHypotheses Q pi) (x0 : X)
    (hS0 : Q.S x0) (hv0 : Q.tau + Q.eps <= Q.v (Q.phi x0)) :
    forall t, Q.S (traj Q pi x0 t)
        /\ Q.tau + Q.eps <= Q.v (Q.phi (traj Q pi x0 t))
        /\ Q.tau <= Q.V (traj Q pi x0 t)
```

the quantized hypotheses of Remark 6.2, which add a reading function and two penalties:

```
  reading           : forall x, Q.S x -> Q.v (Q.phi x) <= Q.v (r x) + a
  model_sensitivity : forall x u, Q.S x ->
                        Q.v (Q.G (r x) u) <= Q.v (Q.G (Q.phi x) u) + a'
  robust_step       : forall x, Q.S x -> Q.tau + Q.eps <= Q.v (r x) + a ->
                        ((Q.tau + Q.eps) + Q.c) + a' <= Q.v (Q.G (r x) (pi (r x)))
```

and the counting lemma's conclusion:

```
theorem data_rate_necessity (A Lam K : Nat) (hA : 1 <= A)
    (u : Nat -> Nat) (hu0 : 1 <= u 0)
    (hrec : forall t, Lam * u t <= u (t + 1) * A)
    (hbound : forall t, u t <= K) : Lam <= A
```

Build and axiom audit: `lake build && lean Check.lean` prints the axioms each theorem depends on. On the checked build (`lean/CHECK_OUTPUT.txt`), the three theorems `reactive_confinement`, `reactive_confinement_quantized` and `fiber_ambiguity` depend on no axioms at all; `predictive_defect_bound`, `witness_bound`, `data_rate_necessity` and `port_necessity` depend on `propext` and `Quot.sound` (through `omega` and `simp`); none depends on `Classical.choice` or `sorryAx`. The Lean tooling is [19]; the formalization was written against Lean `v4.35.0` and does not use Mathlib [66], which would have made the real-valued statements direct.

What is not formalized, and why. Theorem 3.1 and Proposition 3.1 need Lebesgue measure and the volume scaling of linear maps; both exist in Mathlib and neither in core. Conjecture 4.1 is a conjecture. The exhibit is a computation, checked by its own assertions in the Python implementation rather than by proof.

Revision log (October 10 2026)

Changes of substance carry a revision note or a new numbered remark at the point of change. Existing theorem, remark and equation numbers are unchanged; new displays carry primed or lettered tags.

1. §2.1. The grid constrains port values, not the state.
2. Proposition 3.1 (new). Finite-horizon form of Theorem 3.1 with the constant c_0, proved by volume counting; it discharges Remark 3.3 and debt 3 of §11.
3. Remarks 3.5 and 3.6 (new). The rate bound holds at non-equilibrium boundary points, where actuation authority is a separate price; the ϵ and T of Definition 2.1 accounted for.
4. Conjecture 4.1. Restated for equilibria in a boundary collar and for evaluators tied to the landscape.
5. Remark 4.3 (new). The null model violates Assumption 3.2; extensivity is recovered through h_2 at a rate proportional to γ.
6. Remark 6.2 (new), Corollary 6.1, Proposition 10.2, (30). Quantized measurement consumes margin $L_v(1 + L_G)q$; the cap at d holds at sufficient port precision.
7. Definition 10.3, Proposition 10.1. Units stated and a horizon-weighted variant added; the roundwise bound corrected by $c_{0,k}/T_k$.
8. Remark 2.2 (new). Competence floor; the interior retreat as the analogue of the null agent.
9. Remark 10.2 (new). Third public correction, on the 2025 exponential claim; Appendices C and D carry the floored definition.
10. Remark 10.3 (new). What is known about the audit term.
11. Definition 10.4. t^* identified as a first-failure time, the half-life as its median.
12. §8, §9. "measured" → "measurable", the exhibit being unimplemented.
13. Citations. Broken key in the abstract ([P6] → [6]); Remark 4.1 cited; arXiv number for [53]; six entries added.
14. Copy. "Large Languages Models"; the two expansions of LDT reconciled in §5.

Open for the author. (a) Which expansion of LDT is canonical. (b) A source for the "gradient floor" clause of Remark 4.1. (c) The original PDF was set in Latin Modern; this build uses Computer Modern, the only difference in the preamble being the one line that loads the lmodern package.

Revision 3 log (draft, October 14 2026)

[r3: All items below are additions; the October 10 numbering is untouched, new items are lettered, and the markers can be stripped by emptying the `revnote` macro.]

1. Remark 2.A, Table 1. Bits before ports: the bound restated as a bit budget, with the port count tabulated at four precisions and tied to Remark 4.3's $c(\gamma)$ and Remark 6.2's $d\lceil b'/b \rceil$.

2. Lemma 3.A. Cardinality skeleton of Proposition 3.1, display (D1); its second half machine-checked.

3. Remark 4.A. The conjecture's state space: measured Hessian spectra cited; no null model for harness state spaces; Corollary 4.1 read as conditional.

4. Remark 6.A, Appendix E. Lean 4 development (core only) for Proposition 6.1, Remarks 6.1 and 6.2, Lemma 10.1, the pilot witness bound, and Lemma 3.A; axiom audit recorded.

5. §7.1, Table 2. Implementation status of Exhibit 7.1 from the companion repository, with deviations from Appendix A listed and the drift prediction checked.

6. §9.1. Reconciliation with the October 10 revisions of both FRC papers: five correspondences, none a theorem.

7. Remark 10.A. What the checked statements do and do not add.

8. §11. Debts 8 and 9 added; debt 5 half discharged; debt 1 annotated.

9. Citations. Forty-two external entries added, all flagged unverified; the six entries of the October revision carried into the bibliography.

Open for the author (revision 3). (a) Whether Corollary 4.1 should remain in the abstract given Remark 4.A. (b) Whether the shipped exhibit defaults or Appendix A's are canonical. (c) Whether the reconciliation in §9.1 should move into the FRC paper instead, since that paper now cites this one.

References

[1] Ayushi Agarwal. On the formal limits of alignment verification. Preprint, arXiv:2603.08761, 2026.

[2] Ayush Agrawal and Koushil Sreenath. Discrete control barrier functions for safety-critical control of discrete systems with application to bipedal robot navigation. In *Robotics: Science and Systems XIII*, 2017.

[3] Aaron D. Ames, Samuel Coogan, Magnus Egerstedt, Gennaro Notomista, Koushil Sreenath, and Paulo Tabuada. Control barrier functions: Theory and applications. In *2019 18th European Control Conference (ECC)*, pages 3420–3431, 2019.

[4] Jean-Pierre Aubin. *Viability Theory.* Systems & Control: Foundations & Applications. Birkhäuser, Boston, 1991.

[5] Antonio Auffinger, Gérard Ben Arous, and Jiří Černý. Random matrices and complexity of spin glasses. *Communications on Pure and Applied Mathematics*, 66(2):165–201, 2013.

[6] Samuel R. Bowman, Jeeyoon Hyun, Ethan Perez, Edwin Chen, Craig Pettit, Scott Heiner, Kamilė Lukošiūtė, Amanda Askell, Andy Jones, Anna Chen, et al. Measuring progress on scalable oversight for large language models. arXiv:2211.03540, 2022.

[7] Alan J. Bray and David S. Dean. Statistics of critical points of Gaussian fields on large-dimensional spaces. *Physical Review Letters*, 98(15):150201, 2007.

[8] Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, et al. Towards monosemanticity: Decomposing language models with dictionary learning. Transformer Circuits Thread, 2023.

[9] Roger W. Brockett. Minimum attention control. In *Proceedings of the 36th IEEE Conference on Decision and Control*, volume 3, pages 2628–2632, 1997.

[10] Steven L. Brunton, Marko Budišić, Eurika Kaiser, and J. Nathan Kutz. Modern Koopman theory for dynamical systems. *SIAM Review*, 64(2):229–340, 2022.

[11] Collin Burns, Pavel Izmailov, Jan Hendrik Kirchner, Bowen Baker, Leo Gao, Leopold Aschenbrenner, Yining Chen, Adrien Ecoffet, Manas Joglekar, Jan Leike, Ilya Sutskever, and Jeff Wu. Weak-to-strong generalization: Eliciting strong capabilities with weak supervision. arXiv:2312.09390, 2023.

[12] Anna Choromanska, Mikael Henaff, Michaël Mathieu, Gérard Ben Arous, and Yann LeCun. The loss surfaces of multilayer networks. In *Proceedings of the 18th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 38 of *PMLR*, pages 192–204, 2015.

[13] Paul Christiano, Buck Shlegeris, and Dario Amodei. Supervising strong learners by amplifying weak experts. arXiv:1810.08575, 2018.

[14] Fritz Colonius and Christoph Kawan. Invariance entropy for control systems. *SIAM Journal on Control and Optimization*, 48(3):1701–1721, 2009.

[15] Fritz Colonius, Christoph Kawan, and Girish N. Nair. A note on topological feedback entropy and invariance entropy. *Systems & Control Letters*, 62(5):377–381, 2013.

[16] Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. arXiv:2309.08600, 2023.

[17] Yann N. Dauphin, Razvan Pascanu, Caglar Gulcehre, Kyunghyun Cho, Surya Ganguli, and Yoshua Bengio. Identifying and attacking the saddle point problem in high-dimensional non-convex optimization. In *Advances in Neural Information Processing Systems 27*, 2014.

[18] Liam Davis, Leopold Haller, Alberto Alfarano, and Mark Santolucito. Lattice deduction transformers. Preprint, 2026.

[19] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, volume 12699 of *Lecture Notes in Computer Science*, pages 625–635. Springer, 2021.

[20] David S. Dean and Satya N. Majumdar. Large deviations of extreme eigenvalues of random matrices. *Physical Review Letters*, 97(16):160201, 2006.

[21] M. C. F. Donkers, Paulo Tabuada, and W. P. M. H. Heemels. Minimum attention control for linear systems: A linear programming approach. *Discrete Event Dynamic Systems*, 24(2):199–218, 2014.

[22] Patrick Dugan. Control-harness: storyworld evals, MeshEscapePit, oracle control matrix, and the control observability pilot. Software and data, https://github.com/MoralityLabAI/control-harness, revision c7c1b3d (October 11, 2026), 2026.

[23] Patrick Dugan. Towards SuperAlignment intractability proofs: Eigensystem escape and fidelity recursive complexity show lower bounds on corrigibility for RSI agents. Draft, August 15, 2025; revised October 10, 2026. https://moralitylab.xyz/papers/SuperAlignment_Intractability_1.pdf, 2026.

[24] Patrick Dugan. Trade offs between TRM/LDT hybrids. Draft, June 19, 2026. https://moralitylab.xyz/papers/Trade_Offs_between_TRM_LDT_Hybrids__1.pdf, 2026.

[25] Patrick B. Dugan and GPT-5 Thinking. Corrigibility collapse at the limit of FRC: N→N+1 forecasting for RSI super-alignment. Draft, August 25, 2025; revised October 10, 2026. https://moralitylab.xyz/papers/SuperAlignment_Intractability_2.pdf, 2026.

[26] James S. Freudenberg and Douglas P. Looze. Right half plane poles and zeros and design tradeoffs in feedback systems. *IEEE Transactions on Automatic Control*, 30(6):555–565, 1985.

[27] Yan V. Fyodorov. Complexity of random energy landscapes, glass transition, and absolute value of the spectral determinant of random matrices. *Physical Review Letters*, 92(24):240601, 2004.

[28] Yan V. Fyodorov and Ian Williams. Replica symmetry breaking condition exposed by random matrix calculation of landscape complexity. *Journal of Statistical Physics*, 129(5–6):1081–1116, 2007.

[29] Behrooz Ghorbani, Shankar Krishnan, and Ying Xiao. An investigation into neural net optimization via Hessian eigenvalue density. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97 of *PMLR*, pages 2232–2241, 2019.

[30] Antoine Girard and George J. Pappas. Approximation metrics for discrete and continuous systems. *IEEE Transactions on Automatic Control*, 52(5):782–798, 2007.

[31] Ben Goertzel. Toward a formal characterization of real-world general intelligence. In *Proceedings of the 3rd Conference on Artificial General Intelligence (AGI-10)*, pages 74–79. Atlantis Press, 2010.

[32] Oded Goldreich, Shafi Goldwasser, and Dana Ron. Property testing and its connection to learning and approximation. *Journal of the ACM*, 45(4):653–750, 1998.

[33] Ryan Greenblatt, Buck Shlegeris, Kshitij Sachan, and Fabien Roger. AI control: Improving safety despite intentional subversion. arXiv:2312.06942, 2023.

[34] João P. Hespanha, Payam Naghshtabrizi, and Yonggang Xu. A survey of recent results in networked control systems. *Proceedings of the IEEE*, 95(1):138–162, 2007.

[35] Geoffrey Irving, Paul Christiano, and Dario Amodei. AI safety via debate. arXiv:1805.00899, 2018.

[36] Alexia Jolicoeur-Martineau. Less is more: Recursive reasoning with tiny networks. arXiv:2510.04871, 2025.

[37] Leslie Pack Kaelbling, Michael L. Littman, and Anthony R. Cassandra. Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 101(1–2):99–134, 1998.

[38] Christoph Kawan. Invariance entropy of control sets. *SIAM Journal on Control and Optimization*, 49(2):732–751, 2011.

[39] Christoph Kawan. *Invariance Entropy for Deterministic Control Systems: An Introduction*, volume 2089 of *Lecture Notes in Mathematics*. Springer, Cham, 2013.

[40] Mohammad Khajenejad, Matthew Cavorsi, Ruochen Niu, Qiang Shen, and Sze Zheng Yong. Tractable compositions of discrete-time control barrier functions with application to driving safety control. In *2021 European Control Conference (ECC)*, 2021.

[41] Bernard O. Koopman. Hamiltonian systems and transformation in Hilbert space. *Proceedings of the National Academy of Sciences*, 17(5):315–318, 1931.

[42] Pilhwa Lee and Frank C. Park. On the existence and computation of minimum attention optimal control laws. *IEEE Transactions on Automatic Control*, 67(5):2576–2581, 2022.

[43] Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems 36*, 2023.

[44] Ming Li and Paul Vitányi. *An Introduction to Kolmogorov Complexity and Its Applications*. Springer, 4th edition, 2019.

[45] Kevin K. Lin and Fei Lu. Data-driven model reduction, Wiener projections, and the Koopman–Mori–Zwanzig formalism. *Journal of Computational Physics*, 424:109864, 2021.

[46] David Manheim and Scott Garrabrant. Categorizing variants of Goodhart's law. arXiv:1803.04585, 2018.

[47] David Marsh, Liam McAllister, and Timm Wrase. The wasteland of random supergravities. *Journal of High Energy Physics*, 2012(3):102, 2012.

[48] Nuno C. Martins and Munther A. Dahleh. Feedback control in the presence of noisy channels: Bode-like fundamental limitations of performance. *IEEE Transactions on Automatic Control*, 53(7):1604–1615, 2008.

[49] Nuno C. Martins, Munther A. Dahleh, and John C. Doyle. Fundamental limitations of disturbance attenuation in the presence of side information. *IEEE Transactions on Automatic Control*, 52(1):56–66, 2007.

[50] Igor Mezić. Spectral properties of dynamical systems, model reduction and decompositions. *Nonlinear Dynamics*, 41(1–3):309–325, 2005.

[51] C. Mohtadi. Bode's integral theorem for discrete-time systems. *IEE Proceedings D – Control Theory and Applications*, 137(2):57–66, 1990.

[52] Hazime Mori. Transport, collective motion, and Brownian motion. *Progress of Theoretical Physics*, 33(3):423–455, 1965.

[53] Tiberiu Musat. Neural weight norm = Kolmogorov complexity. Preprint, arXiv:2605.10878, 2026.

[54] Girish N. Nair and Robin J. Evans. Stabilization with data-rate-limited feedback: Tightest attainable bounds. *Systems & Control Letters*, 41(1):49–56, 2000.

[55] Girish N. Nair, Robin J. Evans, Iven M. Y. Mareels, and William Moran. Topological feedback entropy and nonlinear stabilization. *IEEE Transactions on Automatic Control*, 49(9):1585–1597, 2004.

[56] Girish N. Nair, Fabio Fagnani, Sandro Zampieri, and Robin J. Evans. Feedback control under data rate constraints: An overview. *Proceedings of the IEEE*, 95(1):108–137, 2007.

[57] Vardan Papyan. Traces of class/cross-class structure pervade deep learning spectra. *Journal of Machine Learning Research*, 21(252):1–64, 2020.

[58] Levent Sagun, Utku Evci, V. Ugur Güney, Yann Dauphin, and Léon Bottou. Empirical analysis of the Hessian of over-parametrized neural networks. arXiv:1706.04454; ICLR 2018 Workshop, 2018.

[59] Anant Sahai and Sanjoy K. Mitter. The necessity and sufficiency of anytime capacity for stabilization of a linear system over a noisy communication link—Part I: Scalar systems. *IEEE Transactions on Information Theory*, 52(8):3369–3395, 2006.

[60] Lee Sharkey, Bilal Chughtai, Joshua Batson, Jack Lindsey, Jeff Wu, Lucius Bushnaq, Nicholas Goldowsky-Dill, Stefan Heimersheim, Alejandro Ortega, Joseph Bloom, et al. Open problems in mechanistic interpretability. arXiv:2501.16496, 2025.

[61] Joar Skalse, Nikolaus H. R. Howe, Dmitrii Krasheninnikov, and David Krueger. Defining and characterizing reward gaming. In *Advances in Neural Information Processing Systems 35*, 2022.

[62] Vishal Srivastava. Fundamental limits of black-box safety evaluation: Information-theoretic and computational barriers from latent context conditioning. Preprint, arXiv:2602.16984, 2026.

[63] Eliran Subag. The complexity of spherical p-spin models—a second moment approach. *The Annals of Probability*, 45(5):3385–3450, 2017.

[64] Paulo Tabuada. *Verification and Control of Hybrid Systems: A Symbolic Approach*. Springer, 2009.

[65] Sekhar Tatikonda and Sanjoy K. Mitter. Control under communication constraints. *IEEE Transactions on Automatic Control*, 49(7):1056–1068, 2004.

[66] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 367–381, 2020.

[67] Mahendra Singh Tomar, Christoph Kawan, and Majid Zamani. Numerical over-approximation of invariance entropy via finite abstractions. *Systems & Control Letters*, 170:105395, 2022.

[68] Alexander Matt Turner, Lisa Thiergart, David Udell, Gavin Leech, Ulisse Mini, and Monte MacDiarmid. Activation addition: Steering language models without optimization. arXiv:2308.10248, 2023.

[69] Guan Wang, Jin Li, Yuhao Sun, Xing Chen, Changling Liu, Yue Wu, Meng Lu, Sen Song, and Yasin Abbasi-Yadkori. Hierarchical reasoning model. arXiv:2506.21734, 2025.

[70] Wing Shing Wong and Roger W. Brockett. Systems with finite communication bandwidth constraints. II. Stabilization with limited information feedback. *IEEE Transactions on Automatic Control*, 44(5):1049–1053, 1999.

[71] Andrew Chi-Chih Yao. Probabilistic computations: Toward a unified measure of complexity. In *18th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 222–227, 1977.

[72] Jasper Yao. The alignment trap: Complexity barriers. Preprint, arXiv:2506.10304, 2025.

[73] Simon Zhuang and Dylan Hadfield-Menell. Consequences of misaligned AI. In *Advances in Neural Information Processing Systems 33*, 2020.

[74] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to AI transparency. arXiv:2310.01405, 2023.

[75] Robert Zwanzig. Memory effects in irreversible thermodynamics. *Physical Review*, 124(4):983–992, 1961.